

## Supplementary Information

### **Splicing-factor expression in cultured human fibroblasts reports proliferative state and the coupling weakens in skin**

[Authors and affiliations to be completed before submission]

This file contains Supplementary Results 1 to 8, Supplementary Figures S1 to S8, Supplementary Tables 1 to 10, Supplementary Notes 1 and 2 and the list of Supplementary Data files. Reference numbers are those of the main reference list. Every number here is reproduced by the scripts named in the Code availability statement.

## Supplementary Results

### **Supplementary Results 1. Division rate under nine treatments and two oxygen tensions (post hoc)**

Division rate was measured in the Cellular Lifespan Study<sup>13,14</sup> rather than manipulated by us, but nine of its treatment arms manipulate it in the same cell lines. Each treated culture was compared with the control culture of the same cell line and oxygen tension nearest in time in culture (224 of 224 treated cultures matched; median difference in days grown 0). The splicing score is the 177-gene score of Fig. 2. The change a treatment makes to the score follows the change it makes to the measured division rate ( $\rho = 0.49$ ,  $P = 4 \times 10^{-15}$ ,  $n = 224$  cultures) and follows the change in the transcriptional proliferation score more closely ( $\rho = 0.79$ ,  $P = 1 \times 10^{-49}$ ; Supplementary Fig. S1a, b). Contact inhibition, the largest manipulation of rate available, lowers the rate by 0.35 divisions per day and the score by 0.73 standard deviations ( $n = 41$  cultures).

Pharmacology separates the two measures of proliferation. With the measured division rate held constant, five of the nine arms keep a residual effect on the splicing score at  $FDR < 0.05$ : dexamethasone +0.53 (95% CI 0.43 to 0.64), oligomycin +0.56 (0.41 to 0.71), mitochondrial nutrient-uptake inhibitors with dexamethasone +0.45 (0.31 to 0.58), oligomycin with dexamethasone +0.41 (0.16 to 0.66) and contact inhibition -0.17 (-0.31 to -0.03); a sixth, galactose, is nominal (+0.15, 95% CI 0.002 to 0.31,  $P = 0.048$ ,  $FDR = 0.07$ ). With the counted rate held constant the four drug treatments among them also carry a higher proliferation score than the rate predicts (dexamethasone +1.27, 1.10 to 1.44; mitochondrial nutrient-uptake inhibitors with dexamethasone +1.08; oligomycin with dexamethasone +0.98; oligomycin +0.54). In the matched differences the proliferation score rises under dexamethasone (+0.81) and under the inhibitors with

dexamethasone (+0.28) and falls under oligomycin (−0.14) and oligomycin with dexamethasone (−0.27). With the proliferation score held constant instead, seven of the nine arms have no residual effect (Supplementary Fig. S1c, Supplementary Table 2); the exceptions are oligomycin, which raises the splicing score above what the proliferation score predicts (+0.28, 0.17 to 0.40), and 2-deoxyglucose, which lowers it (−0.32, −0.43 to −0.20). The proliferation score and the splicing score are strongly correlated with each other, so this second adjustment cannot separate them; what it shows is that these treatments do not move one without the other.

Oxygen tension leaves a further residual. The treatment comparison holds oxygen fixed by design, so it was examined separately among control cultures: fifteen cultures of three cell lines grown at 3% oxygen were matched to control cultures of the same line at 21% oxygen nearest in days grown (median difference 2.3 days). Growth at 3% oxygen lowers the measured division rate by 0.10 divisions per day (12 of 15 cultures lower) and the splicing score by 0.51 (12 of 15 lower), while the proliferation score does not move detectably (−0.26, 10 of 15 lower). In a model of the 104 control libraries with cell line and days grown as covariates, the splicing score is 0.21 lower at 3% oxygen with the measured rate held constant (95% CI −0.39 to −0.03,  $P = 0.022$ ) and 0.33 lower with the proliferation score held constant (−0.44 to −0.23,  $P = 2 \times 10^{-8}$ ). This residual is explained by neither measure of proliferation. It rests on three cell lines, and it counts against a reading in which proliferation is the only determinant of the score.

Apart from contact inhibition and the six cultures given oligomycin with dexamethasone (both −0.35 divisions per day), the arms move the division rate by 0.14 divisions per day or less, so this analysis tests the coupling under perturbation rather than across the full range of division rates. Treatments were not randomised to cell lines and the cultures are serial passages of a few lines. The  $P$  values treat cultures as units; the number of cell lines behind each arm (two to seven; Supplementary Table 2a) and, for oxygen, the per-line differences in Supplementary Table 2d are the conservative reading.

## **Supplementary Results 2. Which genes keep an age association**

If division rate drives the association, individual genes should retain their age association in proportion to how loosely they are coupled to division. In the 107 adult donors, 137 of the 177 genes show an age association and every one of them is a decline. Once a transcriptional proliferation score is included, 19 keep a significant decline and none acquires one (Supplementary Fig. S2c). That last statement holds for the primary cohort only: two genes acquire an association after adjustment in the 133-donor definition and eight in the deposited 142-sample set (Supplementary Table

1). The genes are individually well coupled to proliferation (median  $r = 0.64$ ; Supplementary Fig. S2b), and the share of the age effect a gene keeps falls as that coupling rises ( $\rho = -0.45$ , 95% CI  $-0.56$  to  $-0.32$ ,  $P = 3 \times 10^{-10}$ ,  $n = 177$  genes). A gene's retained share and its correlation with the covariate being adjusted for are not independent quantities, so this relationship is expected in part from the adjustment itself and is not an independent test. The 19 genes that keep a decline are somewhat less tightly coupled than the rest (median 0.49 against 0.65), although the difference is not significant ( $P = 0.16$ ). None of them is one of the twenty factors the ageing literature assays: of the eleven of those factors that belong to the set, none keeps an age association once proliferation is included (Supplementary Fig. S2a).

The set is not carried by its non-spliceosomal members. Restricted to the 127 genes that belong to Reactome mRNA splicing itself, the age effect and its attenuation are the same as for the whole set ( $-0.132$  per decade, 66% removed by adjustment), and so they are for the remaining 50 genes of capping, 3'-end processing and export ( $-0.136$ , 78% removed; Supplementary Table 3); scored in one matrix, the two subsets follow the measured division rate equally closely ( $\rho = 0.66$  and  $0.70$ , whole set  $0.68$  in that matrix; see Supplementary Results 6).

Those 19 genes include core spliceosomal and 3'-processing components — *SF3A3*, *SF3B6*, *SNRPF*, *PPIH*, *PPIL1*, *SNW1*, *CDC5L*, *CWC22*, *CWC27*, *MAGO*H, *PNN*, *HNRNPC*, *RBMX*, *PABPN1*, *ZCRB1*, *CCAR1*, *IK*, *THOC2* and *NUP88*. They are not, however, a stable set. The same analysis repeated under the other cohort definitions keeps 40 genes in the deposited 142-sample set, 56 when only progeria donors are removed, 19 in the primary cohort and none among adults aged 20 to 82 (Fig. 4d, Supplementary Table 1). No gene is therefore common to all four definitions; the three definitions that keep any gene share ten (*IK*, *MAGO*H, *PABPN1*, *PNN*, *PPIH*, *PPIL1*, *SF3A3*, *SF3B6*, *SNRPF* and *SNW1*). In the primary specification a decline remains after adjustment at the level of the whole set, but which genes carry it depends on how the cohort is drawn, and no subset of these genes is recommended here as an alternative index of ageing.

### **Supplementary Results 3. The splicing outcome index carries no age association in the adult cohort**

Splicing-factor abundance and the accuracy of splicing are different quantities, and the tissue evidence for age-associated splicing dysfunction rests on the latter<sup>8</sup>. The donor cohort was reprocessed uniformly with junction-level output<sup>17</sup>, so both could be taken from the same libraries and treated identically.

The outcome measure does not support an ageing interpretation in this cohort. The share of junction-spanning reads falling on junctions absent

from the reference annotation is only weakly related to proliferation ( $r = -0.27$ ,  $n = 107$  donors), and in the 107 adult donors it has no association with age ( $\beta = +1.5 \times 10^{-5}$  per decade, 95% CI  $-3.0 \times 10^{-5}$  to  $+6.0 \times 10^{-5}$ ,  $P = 0.51$ ), before or after adjustment (Supplementary Fig. S3a, b). The same analysis run on all 142 deposited samples gives an age association ( $P = 0.002$ ) that the cohort definition removes: progeria donors and children contribute to it, and the 31 donors aged 83 and over — every one of them from the same cell repository, which has a higher unannotated-junction fraction at every age — carry the remainder. Among adults aged 20 to 82 the estimate runs in the opposite direction ( $P = 0.012$ ). The index also behaves like a measure of library quality rather than of biology: across these donors it falls with the proportion of uniquely mapped reads ( $\rho = -0.43$ ) and rises with multi-mapping ( $\rho = +0.42$ ) and intronic signal ( $\rho = +0.31$ ), and adding those quality measures to the model removes what little age association is left ( $P = 0.96$ ), while the age effect on splicing-factor expression survives them ( $-0.110$  per decade,  $P = 7 \times 10^{-6}$ , and  $-0.039$  after adjustment for proliferation,  $P = 0.003$ ; Supplementary Table 3). The entropy of junction usage behaves the same way. This index was not specified in advance of the analyses in which it first appeared and is reported as exploratory.

Event-level splicing carries the same warning (Supplementary Fig. S3c, d). The number of age-associated events falls from 135, 302, 332 and 98 under the four cohort definitions to 20, 22, 14 and none once proliferation is included. The *COL1A2* event with the strongest age association in the deposited set is flat among adults below 83 (PSI 0.035 to 0.049 by age band;  $\rho = 0.12$ ,  $P = 0.29$ ,  $n = 76$  adults aged 20 to 82; the  $\rho = 0.22$  printed in Supplementary Fig. S3d includes donors under 20) and steps up to 0.072 in the 83-and-over stratum, and the distal junction that produces it is absent from GENCODE v41. Whether splicing outcome deteriorates with age in these cells therefore cannot be decided from this cohort, and the tissue-level evidence for age-associated splicing degeneration<sup>8</sup> stands on its own measurements, not on these.

## Supplementary Results 4. A transcriptome-wide age predictor tracks proliferation (post hoc)

The same reading applies to age predictors built from these cells. A ridge model trained and cross-validated on all expressed genes in the 107 donors predicts chronological age with  $r = 0.72$  (95% CI 0.61 to 0.80; mean absolute error 14 years), far outside a permuted-age null (20 permutations,  $r = -0.31$  to 0.40; Supplementary Table 3). Its predictions track the proliferation score at  $r = -0.74$ , and proliferation accounts for 54% of the variance in predicted age (Supplementary Table 3), while the 20-gene proliferation score by itself predicts age at  $r = 0.46$  (mean absolute error 18 years). A model built after removing the cohort covariates and proliferation

from every gene gives  $r = -0.53$ , but its predictions are nearly constant (57.5 to 59.0 years) and its error equals that of assigning every donor the cohort's mean age (21.6 against 21.5 years), so it carries no information about age. That arm cannot be read as isolating proliferation either: removing the cohort's technical covariates alone already lowers accuracy to  $r = 0.24$ , and repository membership cannot be separated from proliferation in this collection.

## Supplementary Results 5. Reprogramming media and secretomes (post hoc class tests)

Interventions intended to rejuvenate fibroblasts almost all make them divide faster<sup>18</sup>, which makes them the sharpest test of the coupling. Seven reprogramming contrasts — naive and primed media, NHSM, RSeT and 5iLAF formulations<sup>19</sup>, OSK induction in fibroblasts from a 96-year-old donor and the matched arm in which wild-type OCT4 is replaced by the point mutant OCT4YR — and fourteen cell-free secretome preparations applied to recipient fibroblasts were examined on the same terms. One contrast previously counted as reprogramming was reclassified: in that series both the day-3 and the day-7 fibroblast samples are Sendai-OSKM-transduced cells still in fibroblast medium, the switch to reprogramming media having occurred on day 8, so their comparison measures four further days of culture.

Within each class the change in splicing-factor expression rises with the change in the cell cycle (reprogramming media  $\rho = 0.89$ , 95% CI 0.43 to 0.98,  $P = 0.007$ ,  $n = 7$  contrasts from two series; secretomes  $\rho = 0.63$ , 0.14 to 0.87,  $P = 0.017$ ,  $n = 14$  contrasts from eight series; Supplementary Fig. S4b). The within-class slope for reprogramming media is steeper than the slope outside the class (1.68, 95% CI 0.82 to 2.55, against 0.64) and depends on the naive-medium contrast (1.19,  $-0.64$  to 3.02, without it). Predicted from a fit that excludes the class being predicted, 12 of the 21 interventions lie within 0.05 of the prediction and 7 lie more than 0.10 from it (Supplementary Fig. S4a): naive reprogramming medium (+0.34), NHSM (+0.12) and one serum-derived preparation (+0.14) above it, and OSK induction ( $-0.18$ ), the OCT4YR arm ( $-0.14$ ), a trophoblast stem cell secretome ( $-0.20$ ) and a mesenchymal stromal cell secretome ( $-0.13$ ) below it. In naive medium *PNISR*, *LSM14A*, *HNRNPD*, *SRSF7* and *HNRNPA1* rise by 0.91 to 1.44 in  $\log_2$  (Supplementary Fig. S4c). Across the seven reprogramming contrasts *PNISR*, *SRSF2* and *TRA2B* rise in all seven (mean  $\log_2$  0.49 to 0.69), *HNRNPA1* and *HNRNPD* in five, *SRSF1* and *AKAP17A* barely move, and *IMP3* falls (mean  $-0.45$ ).

Across all 21 intervention contrasts the median absolute residual is 0.045 against a median absolute change of 0.111 when each class is predicted from a fit that excludes it (0.038 from the in-sample fit of all 63 contrasts), so about three-fifths of a typical intervention effect is what its effect on the

cell cycle predicts. The share differs by class: about half for secretomes (0.041 against 0.082) and about one-ninth for reprogramming media (0.120 against 0.135), whose individual conditions depart from the line in both directions while their class mean does not (+0.02, 95% CI −0.14 to +0.18). With seven contrasts from two series, five of them sharing one reference arm, that class test could not detect a shift smaller than about 0.22, which is larger than the class's typical effect. The secretome class mean is −0.02 (−0.07 to +0.02,  $n = 14$  contrasts from eight series; detectable shift about 0.07). For secretomes about half of a typical change in splicing-factor transcripts is what the cell-cycle response predicts; for reprogramming media most of it is not, in either direction. A rise in splicing-factor transcripts after an intervention therefore cannot be interpreted without a division-rate measurement alongside the transcriptome.

## **Supplementary Results 6. Which part of the expression machinery lies closest to division (post hoc)**

Splicing is not the only part of the machinery of gene expression that follows the cell cycle, so the splicing set was compared with three Reactome sets scored the same way: Translation, Metabolism of RNA and rRNA processing. The 177-gene set was selected for declining in three datasets, one of which compares late with early passages in the very resource where division rate falls with passage. The comparison was therefore run with two unselected definitions as well — Reactome mRNA Splicing (215 genes) and Processing of Capped Intron-Containing Pre-mRNA (296 genes) — and with the pathway genes lying outside the 177 as a lower bound. No comparison shares a gene: shared genes were removed from the comparator for the selected set and from both sides for the other definitions (removing them from both sides for the selected set as well changes no partial correlation by more than 0.01), except that Metabolism of RNA contains both splicing pathways, so there the splicing genes were removed from the comparator only.

Against the counted division rate (328 libraries) every splicing definition follows the rate more closely than the comparators do (selected set  $\rho = 0.68$ , mRNA Splicing 0.63, pre-mRNA processing 0.65; comparators 0.43 to 0.52). The selected set reaches 0.71 with the score of Fig. 2a; here every set is scored in the expression matrix of the programme audit so that all sets share one filter, and 176 of the 177 genes pass it. Holding a comparator constant leaves the splicing definition with most of its coupling (partial  $\rho = 0.56$  to 0.59 for the selected set, 0.43 to 0.51 for mRNA Splicing, 0.52 to 0.55 for pre-mRNA processing), while holding the splicing definition constant leaves the comparators with none (−0.22 to +0.01; Supplementary Fig. S5a). With the unselected definitions the asymmetry is 66 to 91% of its size with the selected set. It is carried by the selected genes: for the 119

pre-mRNA processing genes outside the 177 it reverses (splicing partial  $-0.07$  to  $+0.15$ , comparators  $0.30$  to  $0.50$ ).

In the 107 donors the same holds for the age effect. Selection inflated only the unadjusted effect ( $-0.133$  per decade for the selected set against  $-0.094$  for mRNA Splicing); the proliferation-adjusted residue is the same ( $-0.040$  and  $-0.042$ ), and with translation also in the model it is  $-0.022$  ( $P = 0.044$ ) and  $-0.023$  ( $P = 0.015$ ), whereas the decline of translation and of Metabolism of RNA disappears once either splicing score is in the model ( $-0.001$  to  $+0.009$ ,  $P = 0.17$  to  $0.94$ ; Supplementary Fig. S5c).

GTE<sub>x</sub> cultures do not reproduce the asymmetry. There proliferation is the 20-marker transcriptional score, not a counted rate, and against it translation lies closer than splicing does: holding the other constant, translation keeps  $0.51$  to  $0.59$  and the splicing definitions  $0.28$  to  $0.41$  ( $n = 511$  cultures; Supplementary Fig. S5b). Against Metabolism of RNA and against rRNA processing the selected set keeps an asymmetry ( $0.60$  against  $-0.29$ , and  $0.53$  against  $0.26$ ) but the unselected definitions do not (differences within their bootstrap intervals). The asymmetry therefore appears where division was counted and in the donor cohort, and not against a transcriptional proliferation score in an independent collection. These scores are strongly correlated with one another ( $r = 0.68$  to  $0.89$  in the counted libraries), so what the comparison can establish is which of them lies closest to division rate, not that they are independently regulated programmes (Supplementary Table 5).

## Supplementary Results 7. Seven senescence panels (post hoc)

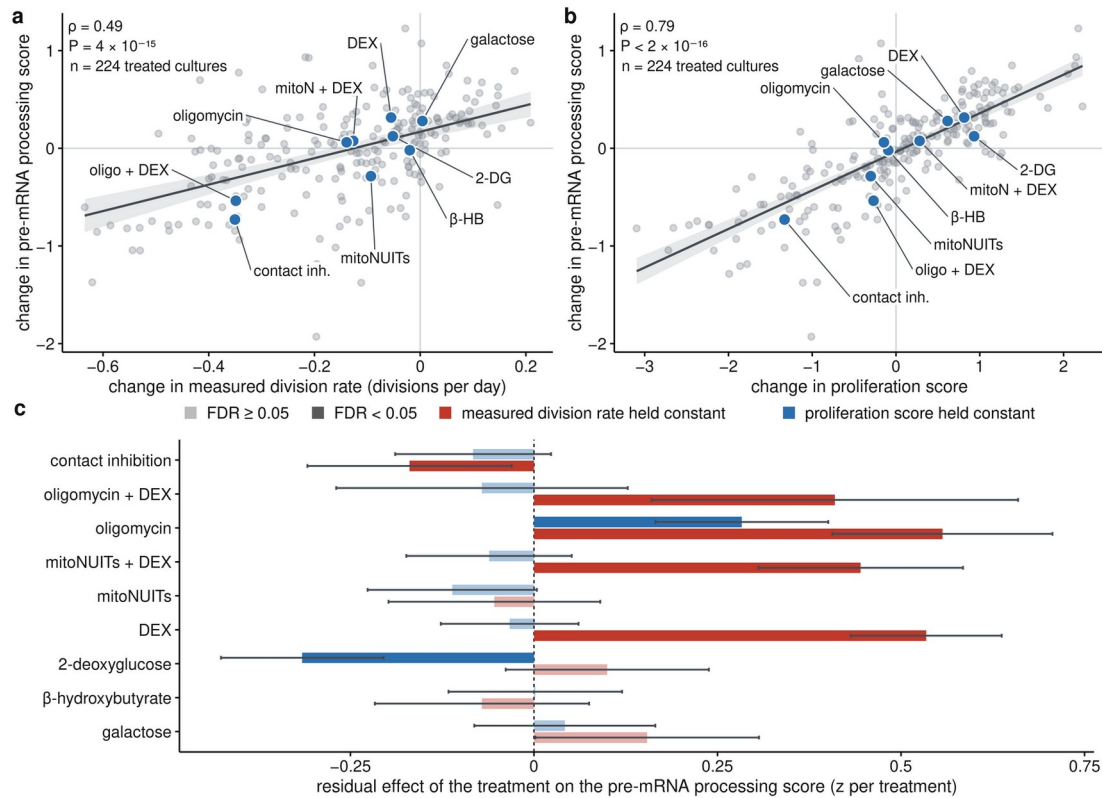
Because the choice of panel decides the answer, five further panels were scored the same way as the Reactome set: SenMayo<sup>22</sup>, the up and down halves of the fibroblast senescence signature of Fridman and Tainsky<sup>36</sup>, and the CellAge genes annotated as inducing or as inhibiting senescence<sup>37</sup>, with the Reactome mitotic cell cycle as the positive control. Across the seven, panels with a larger share of cell-cycle genes tend to correlate more strongly with the measured division rate ( $\rho = 0.71$ , exact two-sided  $P = 0.088$ ,  $n = 7$  panels; Supplementary Fig. S6a); with seven panels the trend does not reach significance. Reactome Cellular Senescence (55% cell-cycle genes) reaches  $\rho = +0.67$  and falls to  $+0.38$  when those genes are removed; SenMayo (1%)  $-0.10$ ; Fridman up (15%)  $-0.07$ ; Fridman down (27%, eleven genes measured)  $+0.38$ , falling to  $+0.01$ ; the CellAge inducers (10%)  $+0.33$ . The CellAge genes annotated as inhibiting senescence do not fit the composition account: 21% of them are cell-cycle genes, yet the panel reaches  $+0.69$  and keeps  $+0.71$  after those genes are removed, as might be expected of genes whose activity lets cells keep dividing.

In the 107 donors, five of the seven panels have an age association before adjustment ( $FDR < 0.05$ ): Reactome Cellular Senescence ( $-0.026$  per decade), the CellAge inhibitors ( $-0.041$ ), Fridman down ( $-0.034$ ) and the cell-cycle control ( $-0.088$ ) decline, and Fridman up rises ( $+0.043$ ). SenMayo and the CellAge inducers have none ( $FDR = 0.72$ ). None of the five keeps its association once proliferation is included; they lose 66 to 100% of the estimate (Supplementary Fig. S6b, Supplementary Table 7).

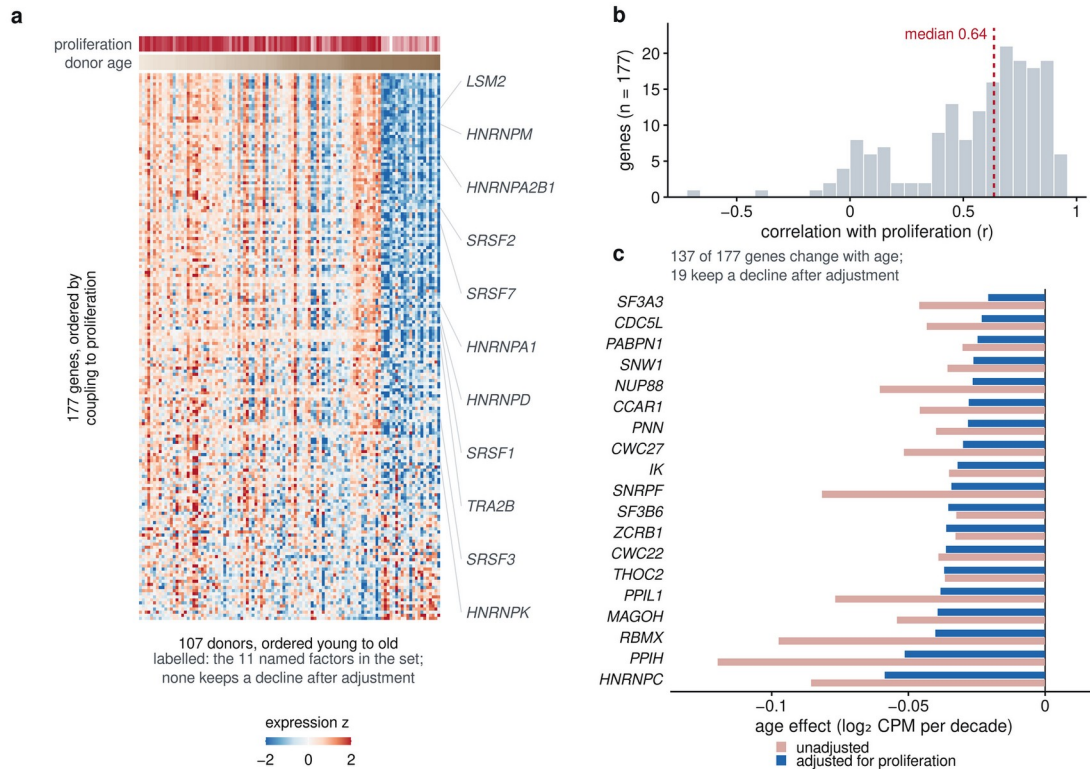
## **Supplementary Results 8. A donor's cultured fibroblasts do not track that donor's skin**

Programme-level state is not shared between a donor's cultured fibroblasts and that donor's skin. Among 388 donors who contributed both cultured fibroblasts and skin, no programme score in the cultured cells was related to the same programme in that donor's skin ( $\rho = -0.10$  to  $0.13$ , none with  $FDR < 0.05$ ; Supplementary Fig. S8c); this concordance was exploratory analysis S3 of the preregistration, for which no success criterion was set (Supplementary Note 1). That is not a failure of the measurement: in the same donors and the same matrices, genes whose expression is strongly genotype-dependent are recovered across the two sample types (*GSTM1*  $\rho = 0.75$ , *ERAP2*  $\rho = 0.75$ , both  $P < 10^{-70}$ ; seven sex-chromosome genes  $0.67$  to  $0.74$ ); this positive control was added post hoc. Programme scores measured in cultured fibroblasts therefore carry no detectable information about the same programmes in the donor's bulk skin ( $|\rho| \leq 0.13$ ,  $n = 388$  donors); because bulk skin is mostly not fibroblasts, this comparison cannot separate the effect of culture from that of cell type.

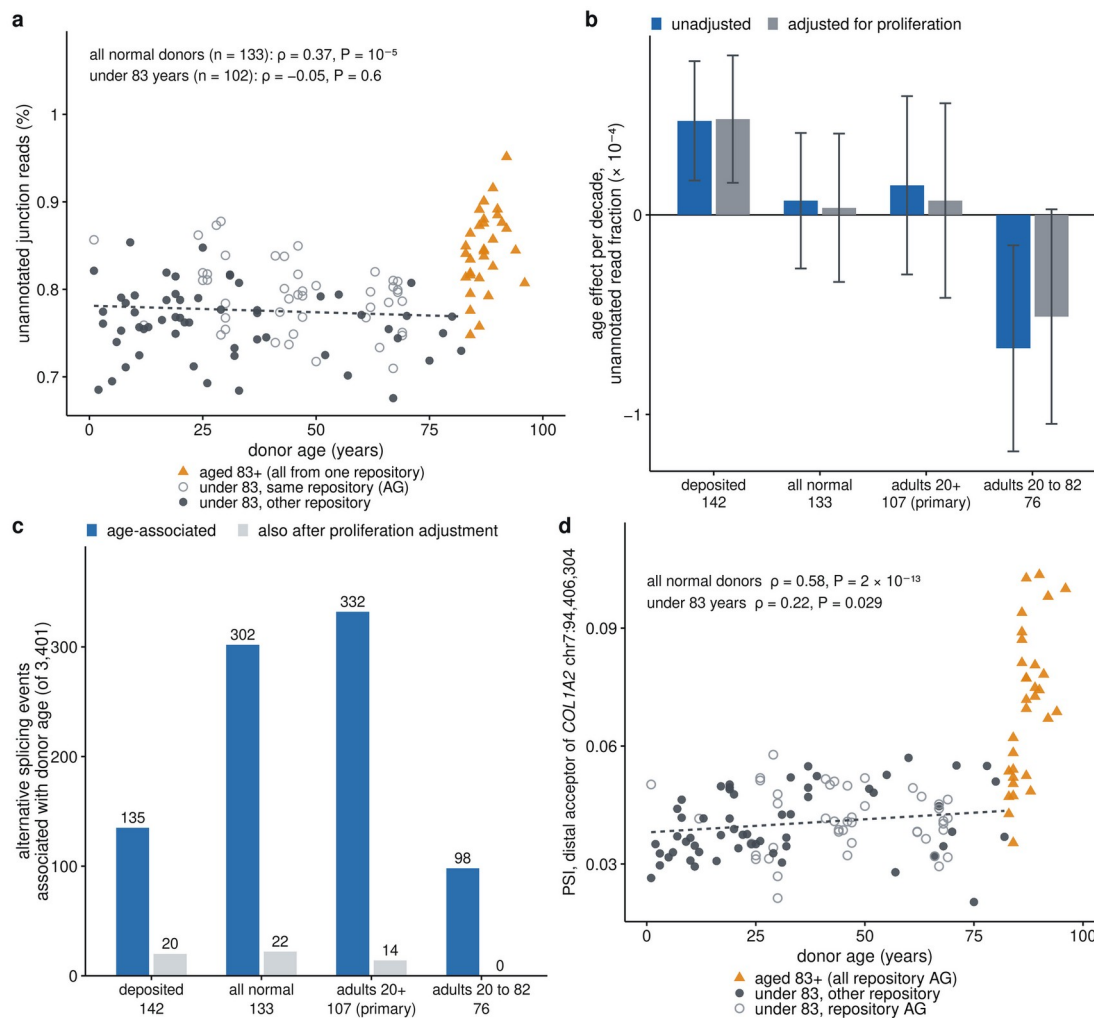
## Supplementary Figures



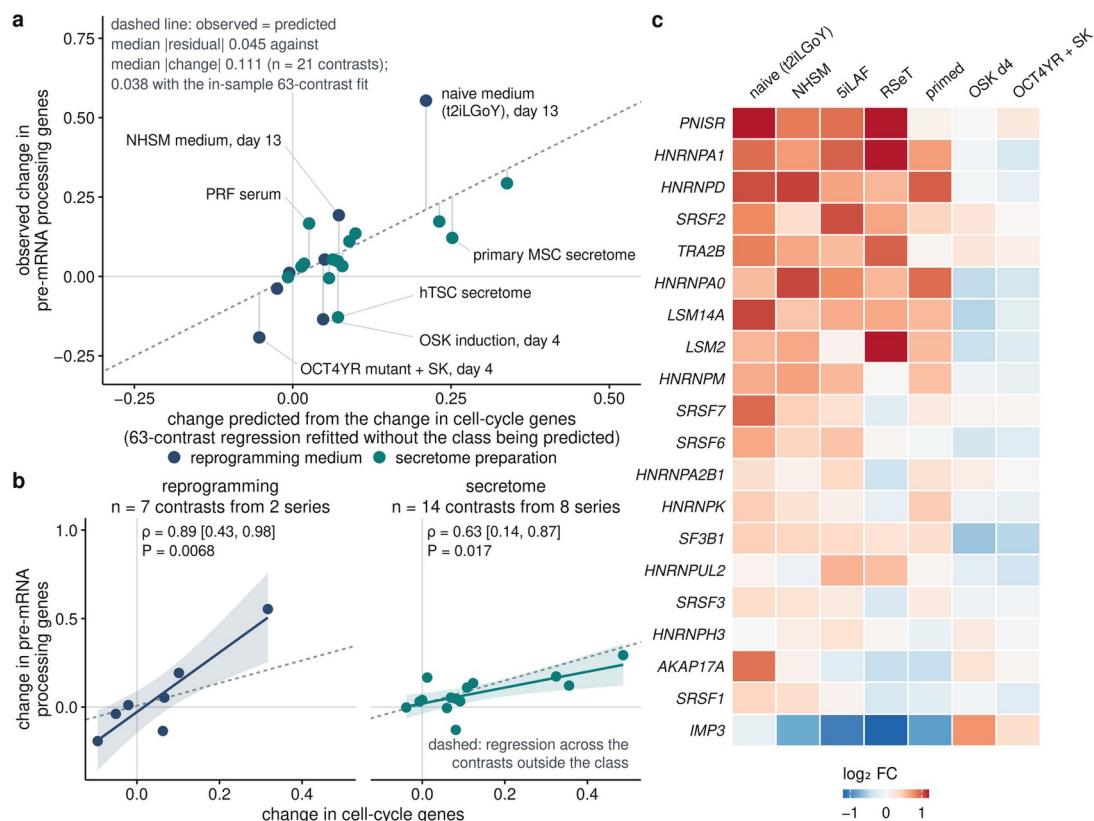
**Supplementary Fig. S1. The coupling under nine perturbations of the same cell lines (post hoc).** (a) For each treated culture, the difference from the control culture of the same cell line and oxygen tension nearest in days grown: change in measured division rate against change in the 177-gene splicing score. Grey, individual cultures; blue, the median of each treatment arm, with the arms discussed in Supplementary Results 1 labelled. (b) The same against the change in the transcriptional proliferation score. (c) Residual effect of each treatment on the splicing score, with cell line, oxygen tension and days grown in the model and either the measured division rate or the proliferation score held constant; bars with 95% confidence intervals, bars at reduced opacity FDR  $\geq 0.05$ , treatments ordered by how much they lower the division rate. Values, and the oxygen contrast, are in Supplementary Table 2.



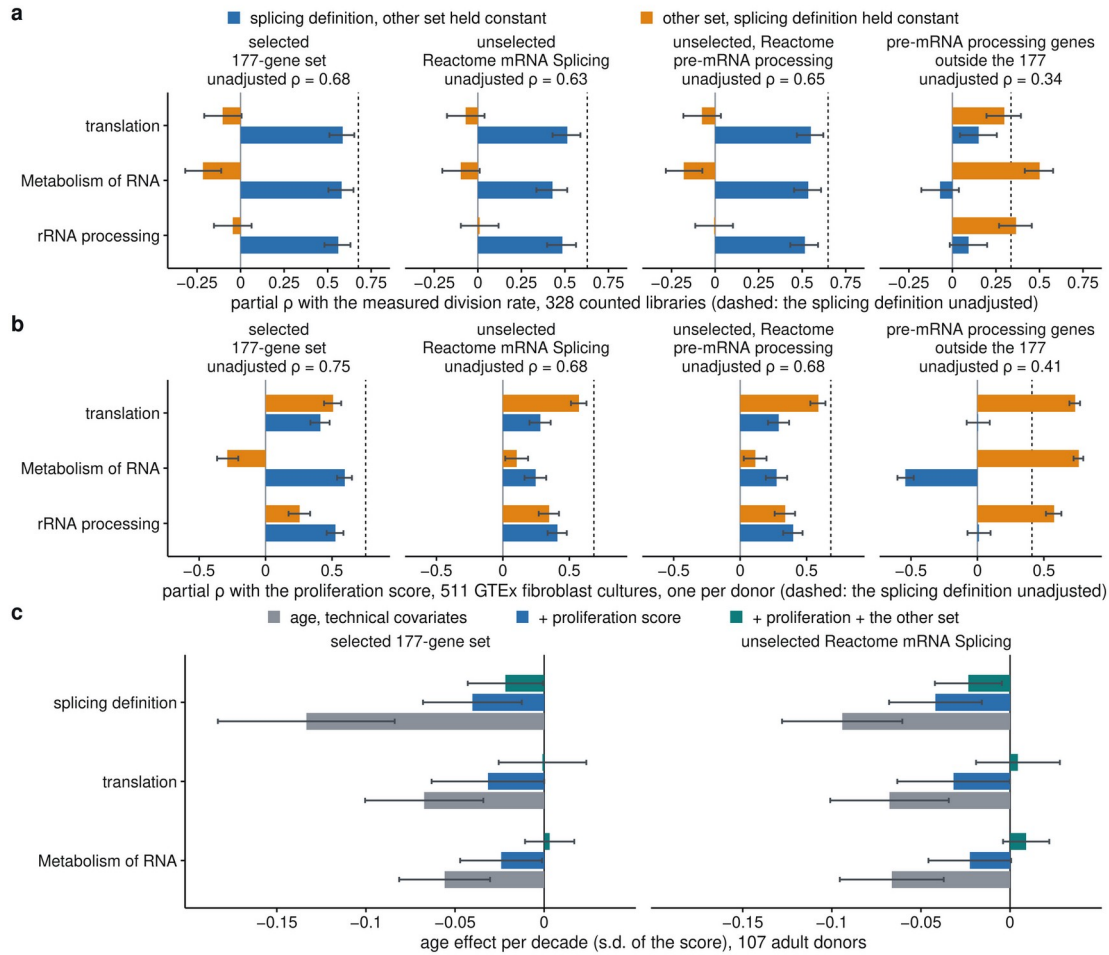
**Supplementary Fig. S2. Which genes keep an age association.** (a) Expression of the 177 genes across the 107 donors ordered young to old, genes ordered by their own correlation with proliferation; bars above show donor age and the proliferation score; the eleven of the twenty named factors that belong to the set are labelled, none of which keeps an age association after adjustment. (b) Distribution of each gene's correlation with the proliferation score. (c) The 19 genes keeping a decline, with the age effect in  $\log_2$  counts per million per decade before (pink) and after (blue) adjustment.



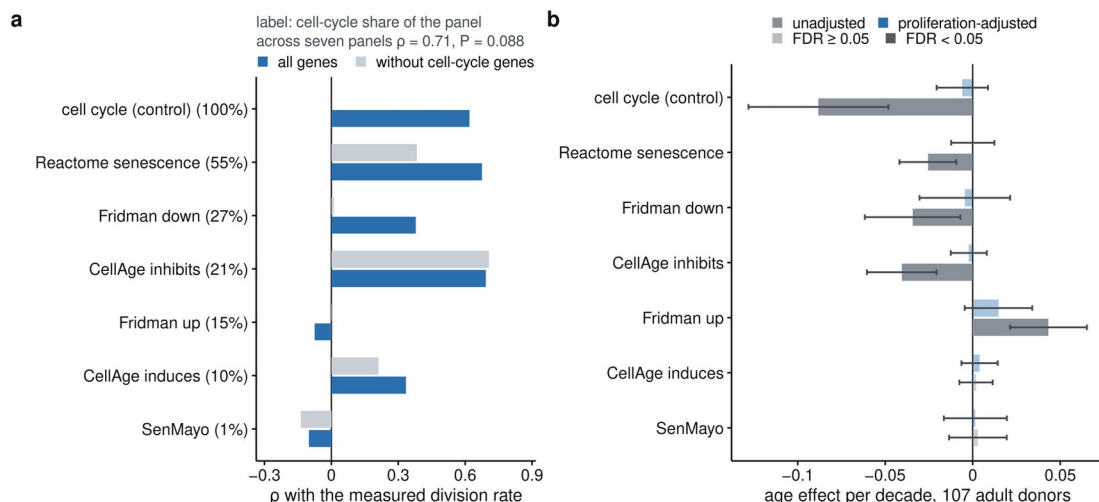
**Supplementary Fig. S3. The splicing outcome index and event-level splicing, neither of which is interpreted.** (a) The share of junction reads on unannotated junctions against donor age in all 133 normal donors, with the 31 donors aged 83 and over — every one from the same cell repository — drawn as triangles and the younger donors of that repository as open symbols. (b) Age effect per decade on the unannotated-read fraction, before and after adjustment for proliferation, under each of the four cohort definitions, as bars with 95% confidence intervals; the deposited set uses sequencing depth as its only covariate. Only the deposited 142-sample set gives a positive age effect, and among adults aged 20 to 82 the estimate runs the other way. (c) Age-associated event counts under each cohort definition, before and after adjustment for proliferation. (d) The *COL1A2* event with the strongest age association in the deposited set against donor age, with the 83-and-over stratum and the cell repository marked; the distal junction that carries that signal is not annotated in GENCODE v41.



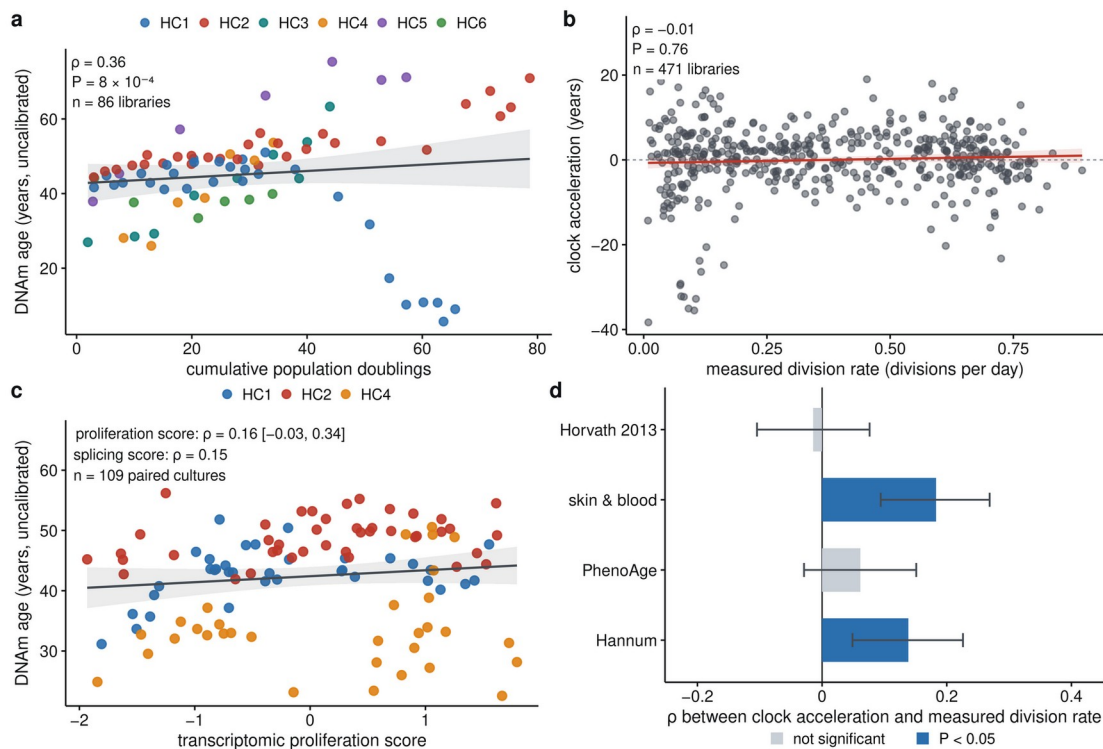
**Supplementary Fig. S4. Reprogramming media and secretomes (post hoc class tests).** (a) Observed change in splicing-factor expression against the change predicted from the cell-cycle response, for seven reprogramming contrasts and fourteen secretome preparations; each class is predicted from the 63-contrast regression refitted without that class. (b) The relationship within each intervention class, with the number of contrasts and of series; dashed line, the regression outside the class. (c) The twenty named splicing factors across the seven reprogramming contrasts, ordered by mean response; grey cells are genes absent from the gene table of that contrast.



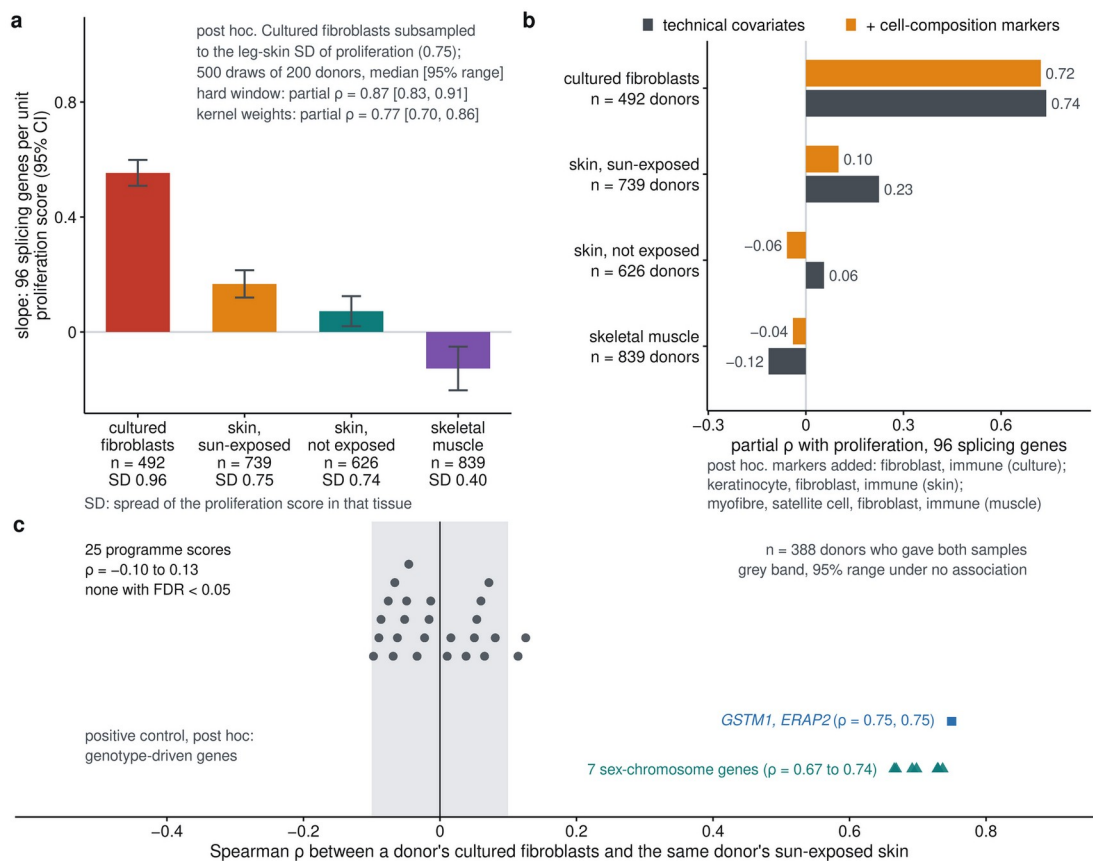
**Supplementary Fig. S5. Which part of the expression machinery lies closest to division (post hoc).** No comparison shares a gene. **(a)** Partial Spearman correlation with the measured division rate across the 328 counted libraries, for the selected 177-gene set, two unselected Reactome definitions and the pathway genes outside the 177, as bars with 95% confidence intervals: the splicing definition with each comparator held constant (blue) and each comparator with the splicing definition held constant (orange); dashed line, the definition's unadjusted correlation in this matrix (0.68 for the 177-gene set, against 0.71 for the score of Fig. 2a; Methods). **(b)** The same comparison against the 20-marker proliferation score in 511 GTEx cultured fibroblasts, where the asymmetry is not reproduced. **(c)** Age effect per decade in the 107 adult donors for the selected set and for unselected Reactome mRNA Splicing, with the cohort's technical covariates, then with the proliferation score, then with the proliferation score and the other set.



**Supplementary Fig. S6. Seven senescence panels (post hoc).** (a) For seven panels scored identically in the 328 counted libraries, the share of measured genes that are also cell-cycle genes printed in the label, and the panel score's correlation with the measured division rate as bars: dark, all genes; light, after every cell-cycle gene is removed; the cell-cycle control has nothing left to score after removal. (b) Age effect per decade for the same panels in the 107 adult donors, before and after adjustment for proliferation, as bars with 95% confidence intervals, bars at reduced opacity  $\text{FDR} \geq 0.05$ ; panels ordered by cell-cycle content.



**Supplementary Fig. S7. Preregistered tests of methylation clocks in the same cultures.** (a) DNAm age from Horvath's multi-tissue clock against the cumulative population doublings of untreated healthy lines at 21% oxygen; absolute values are uncalibrated approximations and only ranks are interpreted. (b) Clock acceleration — DNAm age with days in culture regressed out within each cell line — against the measured division rate across 471 libraries from nine cell lines and nineteen treatments. (c) DNAm age against the transcriptomic proliferation score in the 109 cultures carrying both measurements, with the splicing score's correlation given alongside; colour, cell line. (d) Correlation between clock acceleration and measured division rate for four published clocks, as bars with 95% confidence intervals from Fisher's  $z$ ; blue,  $P < 0.05$ . Supporting comparison, not among the preregistered tests: M2 was fixed for clock acceleration without naming a clock; Horvath's clock, the one the preregistration named unconditionally, carries the recorded verdict, the skin-and-blood clock and PhenoAge were named conditionally on their coefficient tables and each also falls below the preregistered 0.3, and Hannum's clock was added after the preregistration was locked (Supplementary Note 2). M1 (a) met its criterion; M2 (b), M3 (c) and M4 (not drawn; Supplementary Table 8) did not.



**Supplementary Fig. S8. Alternatives to the tissue result, and donor-level concordance.** (a) Post hoc: the regression slope of the splicing score on the proliferation score in each tissue as bars with 95% confidence intervals, the spread (SD) of the proliferation score in each tissue given below, and the partial correlation obtained when cultured fibroblasts are subsampled to the proliferation spread of skin by two schemes (a hard window and kernel weights). (b) Post hoc: the splicing-proliferation partial correlation in each tissue before and after adjustment for cell-composition marker scores (bars, values printed; the markers added in each tissue are listed under the panel). (c) Spearman correlation, across the 388 donors who gave both, between each programme's score in a donor's cultured fibroblasts and in the same donor's skin (preregistered exploratory analysis, no success criterion; Supplementary Note 1), with two genes whose expression is strongly genotype-dependent and seven sex-chromosome genes as a positive control added post hoc.

## Supplementary Tables

### Supplementary Table 1. Cohort sensitivity in the donor series

Every donor-level result under four definitions of the GSE113957 cohort. Age effects are per decade of donor age; "lost" is the proportion of the unadjusted effect removed by adjustment for proliferation.

| cohort                  | n   | rho<br>age-<br>prol | effect | P       | adj<br>effect | adj P   | %<br>lost |
|-------------------------|-----|---------------------|--------|---------|---------------|---------|-----------|
| adults 20+<br>(primary) | 107 | -0.46               | -0.133 | 5.9e-07 | -0.04         | 0.0049  | 70        |
| all normal<br>(133)     | 133 | -0.45               | -0.128 | 4.6e-09 | -0.045        | 0.00044 | 65        |
| adults 20-82            | 76  | -0.25               | -0.151 | 0.00015 | -0.056        | 0.0057  | 63        |
| deposited<br>(142)      | 142 | -0.4                | -0.084 | 3.9e-06 | -0.023        | 0.035   | 73        |

Gene-level counts and the outcome index in the same four cohorts.

| cohort               | unann.<br>effect | unann.<br>P | genes,<br>age | genes<br>kept | PSI<br>events |
|----------------------|------------------|-------------|---------------|---------------|---------------|
| adults 20+ (primary) | 1.5e-05          | 0.51        | 137           | 19            | 332           |
| all normal (133)     | 7.1e-06          | 0.68        | 152           | 56            | 302           |
| adults 20-82         | -6.7e-05         | 0.012       | 107           | 0             | 98            |
| deposited (142)      | 4.7e-05          | 0.0022      | 129           | 40            | 135           |

| cohort | unann.<br>effect | unann.<br>P | genes,<br>age | genes<br>kept | PSI<br>events |
|--------|------------------|-------------|---------------|---------------|---------------|
|--------|------------------|-------------|---------------|---------------|---------------|

## Supplementary Table 2. Treatment perturbation in the counted resource

The splicing score is the 177-gene score of Fig. 2. All of this table is post hoc.

**(a) Treatment arms.** Libraries and cell lines per arm. Arms with fewer than five libraries or present in only one cell line were to be left out of the models; no arm fell below these limits.

| treatment           | libraries | cell lines | modelled |
|---------------------|-----------|------------|----------|
| Control             | 104       | 7          | TRUE     |
| DEX                 | 54        | 7          | TRUE     |
| Contact_Inhibition  | 41        | 3          | TRUE     |
| mitoNUTs+DEX        | 24        | 3          | TRUE     |
| 2-Deoxyglucose      | 24        | 2          | TRUE     |
| mitoNUTs            | 20        | 3          | TRUE     |
| Oligomycin          | 19        | 3          | TRUE     |
| betahydroxybutyrate | 19        | 2          | TRUE     |
| Galactose           | 17        | 2          | TRUE     |
| Oligomycin+DEX      | 6         | 3          | TRUE     |

**(b) Residual effect of each treatment**, with cell line, oxygen tension and days grown in every model and Benjamini-Hochberg correction within each model. A treatment is counted as keeping a residual effect when its coefficient is significant at BH-FDR < 0.05 within the model, and as accounted for by the adjustment named in the first column otherwise. Counts in the text and filled symbols in Supplementary Fig. S1c use BH-FDR < 0.05.

| response<br>and<br>adjustment   | treatment           | effect | 95% CI<br>lower | 95% CI<br>upper | P    | FDR  |
|---------------------------------|---------------------|--------|-----------------|-----------------|------|------|
| splice, rate<br>not in<br>model | 2-<br>Deoxyglucose  | -0.077 | -0.226          | 0.073           | 0.31 | 0.31 |
| splice, rate<br>not in<br>model | betahydroxybutyrate | -0.113 | -0.276          | 0.051           | 0.18 | 0.23 |

| response<br>and<br>adjustment                | treatment               | effect | 95% CI<br>lower | 95% CI<br>upper | P       | FDR     |
|----------------------------------------------|-------------------------|--------|-----------------|-----------------|---------|---------|
| splice, rate<br>not in<br>model              | Contact_In<br>hibition  | −0.530 | −0.659          | −0.400          | 2.5e-14 | 2.3e-13 |
| splice, rate<br>not in<br>model              | DEX                     | 0.401  | 0.291           | 0.512           | 8.8e-12 | 4e-11   |
| splice, rate<br>not in<br>model              | Galactose               | 0.144  | −0.027          | 0.316           | 0.099   | 0.15    |
| splice, rate<br>not in<br>model              | mitoNUITs               | −0.144 | −0.305          | 0.016           | 0.079   | 0.14    |
| splice, rate<br>not in<br>model              | mitoNUITs<br>+DEX       | 0.248  | 0.098           | 0.397           | 0.0013  | 0.0029  |
| splice, rate<br>not in<br>model              | Oligomycin              | 0.384  | 0.221           | 0.547           | 5.9e-06 | 1.8e-05 |
| splice, rate<br>not in<br>model              | Oligomycin<br>+DEX      | 0.164  | −0.110          | 0.438           | 0.24    | 0.27    |
| splice,<br>measured<br>rate held<br>constant | 2-<br>Deoxygluco<br>se  | 0.100  | −0.038          | 0.238           | 0.16    | 0.2     |
| splice,<br>measured<br>rate held<br>constant | betahydrox<br>ybutyrate | −0.071 | −0.217          | 0.075           | 0.34    | 0.39    |
| splice,<br>measured<br>rate held<br>constant | Contact_In<br>hibition  | −0.170 | −0.309          | −0.030          | 0.017   | 0.031   |
| splice,<br>measured<br>rate held<br>constant | DEX                     | 0.534  | 0.432           | 0.637           | 3e-21   | 2.7e-20 |
| splice,<br>measured<br>rate held             | Galactose               | 0.154  | 0.002           | 0.306           | 0.048   | 0.072   |

| response<br>and<br>adjustment                                 | treatment               | effect | 95% CI<br>lower | 95% CI<br>upper | P       | FDR     |
|---------------------------------------------------------------|-------------------------|--------|-----------------|-----------------|---------|---------|
| constant                                                      |                         |        |                 |                 |         |         |
| splice,<br>measured<br>rate held<br>constant                  | mitoNUITs               | −0.054 | −0.198          | 0.090           | 0.46    | 0.46    |
| splice,<br>measured<br>rate held<br>constant                  | mitoNUITs<br>+DEX       | 0.445  | 0.306           | 0.584           | 1.3e-09 | 3.9e-09 |
| splice,<br>measured<br>rate held<br>constant                  | Oligomycin              | 0.556  | 0.406           | 0.706           | 2.9e-12 | 1.3e-11 |
| splice,<br>measured<br>rate held<br>constant                  | Oligomycin<br>+DEX      | 0.410  | 0.160           | 0.659           | 0.0014  | 0.0032  |
| proliferatio<br>n score,<br>measured<br>rate held<br>constant | 2-<br>Deoxygluco<br>se  | 0.876  | 0.646           | 1.106           | 8.9e-13 | 2.7e-12 |
| proliferatio<br>n score,<br>measured<br>rate held<br>constant | betahydrox<br>ybutyrate | −0.203 | −0.445          | 0.040           | 0.1     | 0.12    |
| proliferatio<br>n score,<br>measured<br>rate held<br>constant | Contact_In<br>hibition  | −0.454 | −0.685          | −0.223          | 0.00015 | 0.00022 |
| proliferatio<br>n score,<br>measured<br>rate held<br>constant | DEX                     | 1.270  | 1.099           | 1.441           | 5.5e-37 | 4.9e-36 |
| proliferatio<br>n score,<br>measured                          | Galactose               | 0.262  | 0.009           | 0.516           | 0.043   | 0.055   |

| response<br>and<br>adjustment                                 | treatment               | effect | 95% CI<br>lower | 95% CI<br>upper | P       | FDR     |
|---------------------------------------------------------------|-------------------------|--------|-----------------|-----------------|---------|---------|
| rate held<br>constant                                         |                         |        |                 |                 |         |         |
| proliferatio<br>n score,<br>measured<br>rate held<br>constant | mitoNUITs               | 0.076  | −0.164          | 0.316           | 0.54    | 0.54    |
| proliferatio<br>n score,<br>measured<br>rate held<br>constant | mitoNUITs<br>+DEX       | 1.079  | 0.847           | 1.311           | 9.6e-18 | 4.3e-17 |
| proliferatio<br>n score,<br>measured<br>rate held<br>constant | Oligomycin              | 0.537  | 0.288           | 0.787           | 3.2e-05 | 5.7e-05 |
| proliferatio<br>n score,<br>measured<br>rate held<br>constant | Oligomycin<br>+DEX      | 0.985  | 0.569           | 1.400           | 5e-06   | 1.1e-05 |
| splice,<br>proliferatio<br>n score<br>held<br>constant        | 2-<br>Deoxygluco<br>se  | −0.316 | −0.426          | −0.205          | 5e-08   | 4.5e-07 |
| splice,<br>proliferatio<br>n score<br>held<br>constant        | betahydrox<br>ybutyrate | 0.002  | −0.116          | 0.120           | 0.98    | 0.98    |
| splice,<br>proliferatio<br>n score<br>held<br>constant        | Contact_In<br>hibition  | −0.083 | −0.189          | 0.023           | 0.13    | 0.29    |
| splice,<br>proliferatio<br>n score                            | DEX                     | −0.033 | −0.127          | 0.061           | 0.49    | 0.57    |

| response<br>and<br>adjustment      | treatment          | effect | 95% CI<br>lower | 95% CI<br>upper | P       | FDR     |
|------------------------------------|--------------------|--------|-----------------|-----------------|---------|---------|
| held<br>constant                   |                    |        |                 |                 |         |         |
| splice,<br>proliferatio<br>n score | Galactose          | 0.042  | -0.081          | 0.165           | 0.5     | 0.57    |
| held<br>constant                   |                    |        |                 |                 |         |         |
| splice,<br>proliferatio<br>n score | mitoNUITs          | -0.111 | -0.227          | 0.004           | 0.059   | 0.18    |
| held<br>constant                   |                    |        |                 |                 |         |         |
| splice,<br>proliferatio<br>n score | mitoNUITs<br>+DEX  | -0.061 | -0.174          | 0.052           | 0.29    | 0.52    |
| held<br>constant                   |                    |        |                 |                 |         |         |
| splice,<br>proliferatio<br>n score | Oligomycin         | 0.283  | 0.166           | 0.401           | 3.6e-06 | 1.6e-05 |
| held<br>constant                   |                    |        |                 |                 |         |         |
| splice,<br>proliferatio<br>n score | Oligomycin<br>+DEX | -0.071 | -0.269          | 0.128           | 0.49    | 0.57    |
| held<br>constant                   |                    |        |                 |                 |         |         |

**(c) Matched treated-minus-control differences**, median per arm, against control cultures of the same cell line and oxygen tension nearest in days grown.

| treatment          | cultu<br>res | delta<br>division<br>rate | delta<br>splicing | delta<br>prolifer<br>ation | medi<br>an  <br>delta<br>days |
|--------------------|--------------|---------------------------|-------------------|----------------------------|-------------------------------|
| Contact_Inhibition | 41           | -0.350                    | -0.73             | -1.33                      | 4.9                           |
| Oligomycin+DEX     | 6            | -0.349                    | -0.54             | -0.27                      | 0.0                           |
| Oligomycin         | 19           | -0.139                    | 0.06              | -0.14                      | 0.0                           |
| mitoNUITs+DEX      | 24           | -0.127                    | 0.08              | 0.28                       | 0.0                           |

| treatment           | cultures | delta division rate | delta splicing | delta proliferation | median   delta days |
|---------------------|----------|---------------------|----------------|---------------------|---------------------|
| mitoNUTs            | 20       | -0.093              | -0.29          | -0.30               | 0.0                 |
| DEX                 | 54       | -0.055              | 0.31           | 0.81                | 0.0                 |
| 2-Deoxyglucose      | 24       | -0.052              | 0.12           | 0.93                | 0.0                 |
| betahydroxybutyrate | 19       | -0.020              | -0.02          | -0.09               | 0.0                 |
| Galactose           | 17       | 0.004               | 0.28           | 0.62                | 0.0                 |

**(d) Oxygen tension.** Cultures at 3% oxygen against cultures of the same cell line at 21% oxygen: medians of the differences (3% minus 21%), with the number of cultures in which the 3% value is the lower one. The unmatched rows reproduce a pooled comparison and are given so that its composition can be seen.

| contrast                                                    | line | n at 3% | n at 21% | delta rate | delta splicing | delta proliferation | median   delta day | rate lower | splitting lower |
|-------------------------------------------------------------|------|---------|----------|------------|----------------|---------------------|--------------------|------------|-----------------|
| control, 3% minus 21% oxygen, same line, nearest days grown | all  | 15      | 14       | -0.099     | -0.51          | -0.26               | 2.3                | 12         | 12              |
| control, 3% minus 21% oxygen, same line, nearest days grown | HC 4 | 5       | 4        | -0.019     | -0.51          | 0.04                | 0.4                | 3          | 4               |
| control, 3% minus 21% oxygen, same line, nearest days grown | HC 1 | 5       | 5        | -0.180     | -0.21          | -0.05               | 2.3                | 5          | 3               |
| control, 3% minus 21% oxygen, same line, nearest days grown | HC 2 | 5       | 5        | -0.106     | -0.59          | -0.39               | 2.5                | 4          | 5               |

| contrast                                                                                                                                           | line | n at<br>3% | n at<br>21<br>% | delta<br>rate | delta<br>splici<br>ng | delta<br>prolif<br>eratio<br>n | me<br>dia<br>n   delt<br>a day<br>s | rate<br>low<br>er | spli<br>cin<br>g<br>low<br>er |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|-----------------|---------------|-----------------------|--------------------------------|-------------------------------------|-------------------|-------------------------------|
| control, 3%<br>minus 21%<br>oxygen,<br>matches<br>within 7 days<br>grown only                                                                      | all  | 13         | 13              | -0.099        | -0.51                 | -0.26                          | 2.3                                 | 11                | 11                            |
| contact<br>inhibition,<br>3% minus<br>21% oxygen,<br>same line,<br>nearest days<br>grown                                                           | all  | 20         | 20              | -0.002        | -0.39                 | 0.75                           | 0.0                                 | 15                | 19                            |
| control,<br>unmatched:<br>median of<br>3% minus<br>median of<br>21%, same<br>lines, all days                                                       | all  | 15         | 63              | 0.052         | -0.22                 | 0.77                           |                                     |                   |                               |
| control,<br>unmatched,<br>21% cultures<br>within 7 days<br>of the 3%<br>time span                                                                  | all  | 15         | 38              | -0.047        | -0.52                 | 0.20                           |                                     |                   |                               |
| older<br>sentence<br>(run_division<br>_rate.R):<br>every 3%<br>library minus<br>every 21%<br>library of the<br>same lines,<br>treatments<br>pooled | all  | 35         | 229             | -0.017        | -0.40                 | -0.38                          |                                     |                   |                               |

The oxygen term in models of control cultures, with cell line and days grown as covariates.

| libraries                         | response               | held constant          | n   | n at 3% | effect of 3% oxygen | 95% CI lower | 95% CI upper | P       |
|-----------------------------------|------------------------|------------------------|-----|---------|---------------------|--------------|--------------|---------|
| control cultures, all seven lines | measured division rate | none                   | 104 | 15      | -0.078              | -0.133       | -0.024       | 0.0061  |
| control cultures, all seven lines | proliferation score    | none                   | 104 | 15      | -0.055              | -0.413       | 0.302        | 0.76    |
| control cultures, all seven lines | splicing score         | none                   | 104 | 15      | -0.361              | -0.561       | -0.161       | 0.00064 |
| control cultures, all seven lines | splicing score         | measured division rate | 104 | 15      | -0.213              | -0.392       | -0.034       | 0.022   |
| control cultures, all seven lines | splicing score         | proliferation score    | 104 | 15      | -0.335              | -0.442       | -0.228       | 2e-08   |

| libraries                                                       | response               | held constant          | n   | n at 3% | effect of 3% oxygen | 95% CI lower | 95% CI upper | P      |
|-----------------------------------------------------------------|------------------------|------------------------|-----|---------|---------------------|--------------|--------------|--------|
| lines                                                           |                        |                        |     |         |                     |              |              |        |
| control cultures, all seven lines                               | proliferation score    | measured division rate | 104 | 15      | 0.177               | -0.156       | 0.510        | 0.3    |
| control cultures, the three lines grown at both oxygen tensions | measured division rate | none                   | 78  | 15      | -0.075              | -0.129       | -0.021       | 0.0084 |
| control cultures, the three lines grown at both oxygen tensions | proliferation score    | none                   | 78  | 15      | -0.046              | -0.408       | 0.317        | 0.8    |
| control                                                         | splicing score         | none                   | 78  | 15      | -0.353              | -0.551       | -0.155       | 0.0008 |

| libraries                                                       | response       | held constant          | n  | n at 3% | effect of 3% oxygen | 95% CI lower | 95% CI upper | P     |
|-----------------------------------------------------------------|----------------|------------------------|----|---------|---------------------|--------------|--------------|-------|
| cultures, the three lines grown at both oxygen tensions         |                |                        |    |         |                     |              |              |       |
| control cultures, the three lines grown at both oxygen tensions | splicing score | measured division rate | 78 | 15      | −0.215              | −0.395       | −0.035       | 0.022 |
| control cultures, the three lines grown at both oxygen tensions | splicing score | proliferation score    | 78 | 15      | −0.331              | −0.428       | −0.234       | 4e-09 |

| libraries                                                       | response               | held constant          | n   | n at 3% | effect of 3% oxygen | 95% CI lower | 95% CI upper | P       |
|-----------------------------------------------------------------|------------------------|------------------------|-----|---------|---------------------|--------------|--------------|---------|
| ons                                                             |                        |                        |     |         |                     |              |              |         |
| control cultures, the three lines grown at both oxygen tensions | proliferation score    | measured division rate | 78  | 15      | 0.163               | -0.185       | 0.511        | 0.36    |
| all modelled libraries (the primary treatment model)            | measured division rate | none                   | 328 | 35      | -0.011              | -0.059       | 0.038        | 0.67    |
| all modelled libraries (the primary treatment model)            | proliferation score    | none                   | 328 | 35      | 0.405               | 0.179        | 0.632        | 0.00052 |
| all                                                             | splicin                | none                   | 328 | 35      | -0.317              | -0.453       | -0.182       | 6.3e-06 |

| libraries                                            | response            | held constant          | n   | n at 3% | effect of 3% oxygen | 95% CI lower | 95% CI upper | P       |
|------------------------------------------------------|---------------------|------------------------|-----|---------|---------------------|--------------|--------------|---------|
| modelled libraries (the primary treatment model)     | g score             |                        |     |         |                     |              |              |         |
| all modelled libraries (the primary treatment model) | splicing score      | measured division rate | 328 | 35      | -0.304              | -0.424       | -0.183       | 1.3e-06 |
| all modelled libraries (the primary treatment model) | splicing score      | proliferation score    | 328 | 35      | -0.486              | -0.585       | -0.387       | 2.3e-19 |
| all modelled libraries (the primary treatment model) | proliferation score | measured division rate | 328 | 35      | 0.429               | 0.228        | 0.629        | 3.6e-05 |

| libraries | response | held constant | n | n at 3% | effect of 3% oxygen | 95% CI lower | 95% CI upper | P |
|-----------|----------|---------------|---|---------|---------------------|--------------|--------------|---|
| ary       |          |               |   |         |                     |              |              |   |
| treat     |          |               |   |         |                     |              |              |   |
| ment      |          |               |   |         |                     |              |              |   |
| mod       |          |               |   |         |                     |              |              |   |
| el)       |          |               |   |         |                     |              |              |   |

### Supplementary Table 3. Sensitivity analyses in the donor cohort

**(a) Technical quality covariates.** Uniquely mapped percentage, multi-mapping percentage, intronic signal and exon assignment rate added to the cohort model.

| metric                     | covariates  | beta     | p       | beta adj | p adj  | pct lost |
|----------------------------|-------------|----------|---------|----------|--------|----------|
| splicing-factor expression | cohort      | -0.133   | 5.9e-07 | -0.0402  | 0.0049 | 70       |
| splicing-factor expression | cohort + QC | -0.11    | 7.4e-06 | -0.0386  | 0.0029 | 65       |
| unannotated junction reads | cohort      | 1.48e-05 | 0.51    | 7.19e-06 | 0.77   | -        |
| unannotated junction reads | cohort + QC | 1.08e-06 | 0.96    | 8.87e-06 | 0.7    | -        |
| entropy of junction usage  | cohort      | -4.1e-05 | 0.68    | 0.000156 | 0.1    | -        |
| entropy of junction usage  | cohort + QC | 4.03e-05 | 0.68    | 0.000191 | 0.041  | -        |

**(b) Core spliceosome versus the rest of the set.** The 127 genes belonging to Reactome mRNA splicing, and the 50 genes of capping, 3'-end processing and export.

| metric          | covariates | beta   | p       | beta adj | p adj  | pct lost |
|-----------------|------------|--------|---------|----------|--------|----------|
| splicing-factor | cohort     | -0.133 | 5.9e-07 | -0.040   | 0.0049 | 70       |

| metric                             | covariates | beta   | p       | beta<br>adj | p adj  | pct<br>lost |
|------------------------------------|------------|--------|---------|-------------|--------|-------------|
| expression                         |            |        |         |             |        |             |
| core spliceosome (127 genes)       | cohort     | -0.132 | 4.8e-07 | -0.044      | 0.0043 | 66          |
| capping, 3'-end, export (50 genes) | cohort     | -0.136 | 2.4e-06 | -0.030      | 0.022  | 78          |

The same three scores against the measured division rate in the 328 counted libraries (Spearman correlation with its 95% confidence interval), all scored in the expression matrix of the programme audit.

| gene set                           | genes<br>measured | libraries | rho<br>with<br>division<br>rate | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|------------------------------------|-------------------|-----------|---------------------------------|--------------------|--------------------|---------|
| whole set (177 genes)              | 176               | 328       | 0.68                            | 0.61               | 0.73               | 3.1e-45 |
| core spliceosome (127 genes)       | 126               | 328       | 0.66                            | 0.60               | 0.72               | 9.9e-43 |
| capping, 3'-end, export (50 genes) | 50                | 328       | 0.70                            | 0.64               | 0.75               | 6.4e-49 |

**(c) Base coverage versus read equivalents.** Gene sums divided by read length (51 or 75 nucleotides in this series).

| metric                     | covariates | beta   | p       | beta<br>adj | p adj  | pct<br>lost | scale<br>used       |
|----------------------------|------------|--------|---------|-------------|--------|-------------|---------------------|
| splicing-factor expression | cohort     | -0.133 | 5.9e-07 | -0.040      | 0.0049 | 70          | base<br>coverage    |
| splicing-factor expression | cohort     | -0.134 | 6.7e-07 | -0.039      | 0.0055 | 71          | read<br>equivalents |

**(d) Transcriptome-wide age predictor (post hoc).** Ten-fold cross-validation repeated five times in the 107 donors; the permuted-age null is twenty repetitions.

| model                                   | r     | 95%<br>CI<br>lower | 95%<br>CI<br>upper | MAE  | r with<br>proliferation<br>score |
|-----------------------------------------|-------|--------------------|--------------------|------|----------------------------------|
| all genes                               | 0.72  | 0.61               | 0.80               | 14.2 | −0.74                            |
| all genes, cohort<br>covariates removed | 0.24  | 0.06               | 0.42               | 21.2 | −0.75                            |
| proliferation score alone               | 0.46  | 0.29               | 0.59               | 18.5 | −1.00                            |
| all genes, proliferation<br>removed     | −0.53 | −0.66              | −0.38              | 21.6 | 0.25                             |
| permuted age (one<br>draw)              | −0.28 | −0.45              | −0.10              | 21.9 | 0.44                             |

**(e) Negative control for the proliferation adjustment.** The unadjusted and the proliferation-adjusted cohort models applied to 1,000 random gene sets matched to the splicing set for expression level and containing no splicing or cell-cycle genes. The share removed is given for the random sets that have an age association of their own, because it is undefined without one; effects are in the units of the splicing score.

| quantity                                                                        | value            |
|---------------------------------------------------------------------------------|------------------|
| random gene sets drawn                                                          | 1000             |
| sets with an age association of their own<br>(unadjusted $P < 0.05$ )           | 205              |
| of those, increasing with age / decreasing with<br>age                          | 184 / 21         |
| share of their age effect removed by<br>adjustment, %: median                   | 56               |
| the same, interquartile range                                                   | 43 to 71         |
| the same, 2.5th to 97.5th percentile                                            | 13 to 98         |
| of those, sets losing at least as much as the<br>splicing set (69.8%)           | 55 of 205 (27%)  |
| of those, sets keeping an age effect after<br>adjustment (adjusted $P < 0.05$ ) | 10 of 205        |
| unadjusted age effect per decade, range over<br>all sets                        | −0.018 to +0.030 |
| smallest unadjusted P over all sets                                             | 7.2e-05          |
| smallest adjusted P over all sets                                               | 0.01             |
| splicing set, unadjusted age effect per decade<br>(P)                           | −0.133 (5.9e-07) |
| splicing set, adjusted age effect per decade (P)                                | −0.040 (0.0049)  |
| random sets with an unadjusted effect at least                                  | 0                |

| quantity                                           | value |
|----------------------------------------------------|-------|
| as large as the splicing set's                     |       |
| random sets with an unadjusted P at least as small | 0     |
| random sets with an adjusted P at least as small   | 0     |

**(f) Which technical covariate the residual depends on.** The proliferation-adjusted age effect of the 177-gene score in the 107 donors, per decade with its 95% confidence interval, under subsets of the cohort's technical covariates; the share removed is relative to the unadjusted model that carries the same covariates. The two technical terms overlap: 13 of the 107 libraries were sequenced on the HiSeq 2500, all of them from the AG repository.

| covariates                                           | unadjusted | P       | adjusted | 95% CI (adjusted) | P (adj) | % removed |
|------------------------------------------------------|------------|---------|----------|-------------------|---------|-----------|
| depth                                                | -0.099     | 6.6e-05 | -0.022   | -0.049 to 0.005   | 0.11    | 78        |
| depth + repository                                   | -0.116     | 3.3e-06 | -0.038   | -0.063 to -0.012  | 0.0038  | 68        |
| depth + instrument                                   | -0.101     | 8.7e-05 | -0.015   | -0.043 to 0.013   | 0.29    | 85        |
| depth + sex                                          | -0.105     | 2.6e-05 | -0.025   | -0.053 to 0.002   | 0.067   | 76        |
| depth + repository + instrument + sex (cohort model) | -0.133     | 5.9e-07 | -0.040   | -0.068 to -0.013  | 0.0049  | 70        |

**(f) The age effect as a mediation model (post hoc).** Age as the exposure, the proliferation score as the mediator, the cohort's technical covariates in every equation, and a percentile bootstrap over donors. The mediator is a proxy for the counted division rate, so the indirect path is attenuated by its reliability; the second table gives the proportion mediated under a range of reliabilities.

| quantity | estimate | 95% CI lower | 95% CI upper |
|----------|----------|--------------|--------------|
| total    | -0.2149  |              |              |
| a        | -0.1828  |              |              |

| quantity | estimate | 95% CI lower | 95% CI upper |
|----------|----------|--------------|--------------|
| b        | 0.8212   |              |              |
| direct   | −0.0648  | −0.1058      | −0.0292      |
| indirect | −0.1501  | −0.2083      | −0.0919      |
| prop     | 0.6985   | 0.5366       | 0.8546       |

| reliability | proportion mediated |
|-------------|---------------------|
| 1.00        | 0.70                |
| 0.80        | 0.87                |
| 0.60        | 1.00                |
| 0.39        | 1.00                |

### Supplementary Table 4. Datasets and contrasts

All 63 contrasts, with the class assigned by experimental manipulation, the change in the cell-cycle and splicing sets, and the source publication of each series. The secretome class holds 14 contrasts.

| accession     | class                    | contrast                                   | cell<br>cycle | splicing |
|---------------|--------------------------|--------------------------------------------|---------------|----------|
| GSE14969<br>4 | culture and<br>metabolic | Fibroblast-D7 vs<br>Fibroblast-D3          | 0.097         | 0.083    |
| GSE17984<br>8 | culture and<br>metabolic | 2-Deoxyglucose vs<br>Control (21% O2)      | 0.063         | −0.05    |
| GSE17984<br>8 | culture and<br>metabolic | betahydroxybutyrate<br>vs Control (21% O2) | −0.125        | −0.038   |
| GSE17984<br>8 | culture and<br>metabolic | Contact_Inhibition vs<br>Control (21% O2)  | −0.51         | −0.184   |
| GSE17984<br>8 | culture and<br>metabolic | DEX vs Control (21%<br>O2)                 | 0.281         | 0.142    |
| GSE17984<br>8 | culture and<br>metabolic | Galactose vs Control<br>(21% O2)           | 0.062         | 0.044    |
| GSE17984<br>8 | culture and<br>metabolic | 3% vs 21% oxygen<br>(Control)              | −0.074        | −0.127   |
| GSE17984<br>8 | culture and<br>metabolic | mitoNUIs vs<br>Control (21% O2)            | −0.021        | −0.037   |
| GSE17984<br>8 | culture and<br>metabolic | mitoNUIs+DEX vs<br>Control (21% O2)        | 0.239         | 0.105    |
| GSE17984<br>8 | culture and<br>metabolic | Oligomycin vs<br>Control (21% O2)          | 0.097         | 0.108    |

| accession     | class                    | contrast                                     | cell<br>cycle | splicing |
|---------------|--------------------------|----------------------------------------------|---------------|----------|
| GSE17984<br>8 | culture and<br>metabolic | Oligomycin+DEX vs<br>Control (21% O2)        | 0.129         | 0.065    |
| GSE17984<br>8 | culture and<br>metabolic | SURF1 mutation vs<br>normal (Control)        | -0.334        | -0.154   |
| GSE11395<br>7 | donor age                | donor age, per<br>decade                     | -0.085        | -0.054   |
| GSE30737<br>7 | donor age                | old vs young dermal<br>fibroblast            | -0.324        | -0.178   |
| GSE11696<br>8 | photoprotection          | prered_vs_uv 1h                              | -0.084        | 0.008    |
| GSE11696<br>8 | photoprotection          | prered_vs_uv 4h                              | -0.061        | 0        |
| GSE11696<br>8 | photoprotection          | Pre-Red vs UV,<br>injury-residualised,<br>1h | -0.085        | 0.002    |
| GSE11696<br>8 | photoprotection          | Pre-Red vs UV,<br>injury-residualised,<br>4h | -0.063        | -0.002   |
| GSE11696<br>8 | photoprotection          | red_vs_control 1h                            | -0.063        | -0.03    |
| GSE11696<br>8 | photoprotection          | red_vs_control 4h                            | -0.058        | -0.002   |
| GSE22241<br>4 | photoprotection          | UVB + TAT-UIFSP5<br>vs UVB (rescue)          | -0.121        | -0.012   |
| GSE22241<br>4 | photoprotection          | UVB + TAT-UIFSP6<br>vs UVB (rescue)          | -0.027        | 0.048    |
| GSE24048<br>6 | photoprotection          | 0.06% Osmoter<br>alone, no UV                | -0.101        | -0.101   |
| GSE24048<br>6 | photoprotection          | 0.02% Osmoter<br>alone, no UV                | 0.001         | -0.016   |
| GSE24048<br>6 | photoprotection          | UV + 0.06%<br>Osmoter vs UV<br>(rescue)      | 0.076         | 0.024    |
| GSE24048<br>6 | photoprotection          | UV + 0.02%<br>Osmoter vs UV<br>(rescue)      | 0.238         | 0.103    |
| GSE32695<br>1 | photoprotection          | UVB + sauchinone<br>vs UVB (rescue)          | -0.035        | 0.175    |
| GSE14969      | reprogramming            | 5iLAF-D13 vs                                 | 0.066         | 0.053    |

| accession     | class                  | contrast                                                       | cell<br>cycle | splicing |
|---------------|------------------------|----------------------------------------------------------------|---------------|----------|
| 4             | media                  | Fibroblast-D3                                                  |               |          |
| GSE14969<br>4 | reprogramming<br>media | NHSM-D13 vs<br>Fibroblast-D3                                   | 0.102         | 0.193    |
| GSE14969<br>4 | reprogramming<br>media | Primed-D13 vs<br>Fibroblast-D3                                 | -0.021        | 0.011    |
| GSE14969<br>4 | reprogramming<br>media | RSeT-D13 vs<br>Fibroblast-D3                                   | -0.051        | -0.038   |
| GSE14969<br>4 | reprogramming<br>media | t2iLGoY-D13 vs<br>Fibroblast-D3                                | 0.317         | 0.554    |
| GSE29723<br>3 | reprogramming<br>media | O4YRSK +dox vs -<br>dox, day 4                                 | -0.095        | -0.192   |
| GSE29723<br>3 | reprogramming<br>media | OSK +dox vs -dox,<br>day 4                                     | 0.063         | -0.135   |
| GSE13956<br>3 | secretome              | senescent-cell CM,<br>day 4 vs day 0                           | 0.092         | 0.033    |
| GSE13956<br>3 | secretome              | senescent-cell CM,<br>day 10 vs day 0                          | -0.039        | -0.002   |
| GSE25180<br>7 | secretome              | BM-MSC<br>conditioned medium<br>vs DMEM                        | 0.109         | 0.111    |
| GSE25180<br>7 | secretome              | BM-MSC EV-<br>depleted medium vs<br>DMEM                       | 0.123         | 0.135    |
| GSE25180<br>7 | secretome              | BM-MSC small EVs<br>vs DMEM                                    | 0             | 0.04     |
| GSE25180<br>7 | secretome              | BM-MSC non-small<br>EVs vs DMEM                                | -0.006        | 0.031    |
| GSE26605<br>2 | secretome              | cytokine-primed<br>hMSC-CM vs<br>untreated (both TGF-<br>beta) | 0.081         | 0.047    |
| GSE26605<br>2 | secretome              | resting hMSC-CM vs<br>untreated (both TGF-<br>beta)            | 0.069         | 0.053    |
| GSE26824<br>8 | secretome              | PRF serum vs<br>untreated (gingival,<br>boundary)              | 0.012         | 0.167    |
| GSE27980<br>4 | secretome              | ADSC nanovesicles<br>vs control                                | 0.324         | 0.173    |

| accession     | class      | contrast                                    | cell<br>cycle | splicing |
|---------------|------------|---------------------------------------------|---------------|----------|
| GSE28205<br>4 | secretome  | hTSC secretome vs<br>ES-CM                  | 0.081         | −0.129   |
| GSE29318<br>6 | secretome  | endothelial-cell EVs<br>vs control, 72 h    | 0.487         | 0.293    |
| GSE30674<br>8 | secretome  | primary MSC<br>secretome vs control         | 0.356         | 0.121    |
| GSE30674<br>8 | secretome  | immortalised MSC<br>secretome vs control    | 0.06          | −0.006   |
| GSE10970<br>0 | senescence | deep replicative<br>senescence              | −1.906        | −1.448   |
| GSE10970<br>0 | senescence | early replicative<br>senescence             | −1.833        | −1.047   |
| GSE17984<br>8 | senescence | late vs early passage                       | −0.359        | −0.252   |
| GSE19105<br>5 | senescence | P27 vs P4                                   | −0.233        | −0.21    |
| GSE30695<br>7 | senescence | DDX58_KO<br>senescent vs<br>proliferating   | −1.21         | −0.524   |
| GSE30695<br>7 | senescence | IFIH1_KO senescent<br>vs proliferating      | −1.373        | −0.924   |
| GSE30695<br>7 | senescence | pLenti senescent vs<br>proliferating        | −1.297        | −0.888   |
| GSE30695<br>7 | senescence | siScramble<br>senescent vs<br>proliferating | −1.073        | −0.746   |
| GSE93535      | senescence | SIPS vs quiescent                           | −0.666        | −0.409   |
| GSE11696<br>8 | UV injury  | uv_vs_control 1h                            | −0.005        | −0.016   |
| GSE11696<br>8 | UV injury  | uv_vs_control 4h                            | −0.084        | −0.082   |
| GSE13453<br>3 | UV injury  | HDF UVB vs no UV<br>(2 donors)              | 0.06          | 0.103    |
| GSE22241<br>4 | UV injury  | UVB vs control<br>(injury)                  | 0.133         | 0.112    |
| GSE24048<br>6 | UV injury  | UV vs control<br>(injury)                   | −0.089        | −0.052   |
| GSE32947<br>5 | UV injury  | UVA, 3 consecutive<br>days vs NC (injury)   | 0.799         | 0.548    |

Source publication of each series.

| accession | source publication                                                                                                                    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|
| GSE149694 | Liu, X. et al. Nature 586, 101-107 (2020)                                                                                             |
| GSE179848 | Sturm, G. et al. Sci. Data 9, 751 (2022); Commun. Biol. 6, 22 (2023)                                                                  |
| GSE113957 | Fleischer, J. G. et al. Genome Biol. 19, 221 (2018)                                                                                   |
| GSE307377 | Tanaka, H. et al. bioRxiv<br><a href="https://doi.org/10.1101/2025.09.17.676776">https://doi.org/10.1101/2025.09.17.676776</a> (2025) |
| GSE116968 | no associated publication                                                                                                             |
| GSE222414 | Yang, T. et al. MedComm 5, e625 (2024)                                                                                                |
| GSE240486 | no associated publication                                                                                                             |
| GSE326951 | Zhang, X. et al. Pharm. Biol. 64, 764-782 (2026)                                                                                      |
| GSE297233 | Lu, J. Y. et al. Cell 188, 5895-5911.e17 (2025)                                                                                       |
| GSE139563 | Lopez-Antona, I. et al. Aging Cell 21, e13580 (2022)                                                                                  |
| GSE251807 | Lei, R., da Silva, T. B., Cui, Z. & Ye, H. Sci. Rep. 15, 19383 (2025)                                                                 |
| GSE266052 | Brizio, M. et al. Stem Cell Res. Ther. 15, 329 (2024)                                                                                 |
| GSE268248 | Imani, A., Panahipour, L., Kuhtreiber, H., Mildner, M. & Gruber, R. Cells 13, 1308 (2024)                                             |
| GSE279804 | Yao, T. et al. J. Cell. Mol. Med. 29, e70877 (2025)                                                                                   |
| GSE282054 | Abdelmohsen, K. et al. Aging Cell 25, e70368 (2026)                                                                                   |
| GSE293186 | Yuan, H. et al. J. Invest. Dermatol. 146, 1369-1381.e11 (2026)                                                                        |
| GSE306748 | no associated publication                                                                                                             |
| GSE109700 | De Cecco, M. et al. Nature 566, 73-78 (2019)                                                                                          |
| GSE191055 | Hasegawa, T. et al. Cell 186, 1417-1431.e20 (2023)                                                                                    |
| GSE306957 | Victorelli, S. et al. Nat. Commun. 16, 10992 (2025)                                                                                   |
| GSE93535  | Lammermann, I. et al. npj Aging Mech. Dis. 4, 4 (2018)                                                                                |
| GSE134533 | no associated publication                                                                                                             |
| GSE329475 | Zhong, J. et al. Cells 15, 1497 (2026)                                                                                                |

**Intervention classes predicted out of sample (post hoc).** Median absolute residual against median absolute change, from the in-sample fit of all 63 contrasts and from fits that exclude the class being predicted.

| scheme                                            | group                | con<br>tras<br>ts | seri<br>es | media<br>n  <br>chang<br>e | media<br>n  <br>residu<br>al | shar<br>e<br>predi<br>cted | wit<br>hin<br>0.0<br>5 | bey<br>ond<br>0.1<br>0 |
|---------------------------------------------------|----------------------|-------------------|------------|----------------------------|------------------------------|----------------------------|------------------------|------------------------|
| in-sample<br>(63-contrast<br>fit)                 | all<br>interventions | 21                | 10         | 0.111                      | 0.038                        | 0.65                       | 13                     | 7                      |
| in-sample<br>(63-contrast<br>fit)                 | reprogramming        | 7                 | 2          | 0.135                      | 0.115                        | 0.14                       | 3                      | 4                      |
| in-sample<br>(63-contrast<br>fit)                 | secretome            | 14                | 8          | 0.082                      | 0.036                        | 0.56                       | 10                     | 3                      |
| leave-class-<br>out                               | all<br>interventions | 21                | 10         | 0.111                      | 0.045                        | 0.59                       | 12                     | 7                      |
| leave-class-<br>out                               | reprogramming        | 7                 | 2          | 0.135                      | 0.120                        | 0.11                       | 3                      | 4                      |
| leave-class-<br>out                               | secretome            | 14                | 8          | 0.082                      | 0.041                        | 0.50                       | 9                      | 3                      |
| all 21<br>interventions<br>left out               | all<br>interventions | 21                | 10         | 0.111                      | 0.041                        | 0.63                       | 12                     | 7                      |
| all 21<br>interventions<br>left out               | reprogramming        | 7                 | 2          | 0.135                      | 0.113                        | 0.16                       | 3                      | 4                      |
| all 21<br>interventions<br>left out               | secretome            | 14                | 8          | 0.082                      | 0.038                        | 0.53                       | 9                      | 3                      |
| class and<br>same-series<br>contrasts<br>left out | all<br>interventions | 21                | 10         | 0.111                      | 0.045                        | 0.59                       | 12                     | 7                      |
| class and<br>same-series<br>contrasts<br>left out | reprogramming        | 7                 | 2          | 0.135                      | 0.120                        | 0.11                       | 3                      | 4                      |
| class and<br>same-series<br>contrasts<br>left out | secretome            | 14                | 8          | 0.082                      | 0.041                        | 0.50                       | 9                      | 3                      |

Mean class residual with its 95% confidence interval and the shift detectable with 80% power.

| scheme                                        | class                 | unit     | n  | mean residual | 95% CI lower | 95% CI upper | detectable shift | class median   change |
|-----------------------------------------------|-----------------------|----------|----|---------------|--------------|--------------|------------------|-----------------------|
| in-samp<br>le<br>(63-<br>contr<br>ast<br>fit) | repro<br>gram<br>ming | contrast | 7  | 0.017         | -0.143       | 0.177        | 0.221            | 0.135                 |
| in-samp<br>le<br>(63-<br>contr<br>ast<br>fit) | repro<br>gram<br>ming | series   | 2  | -0.038        | -1.652       | 1.577        | 2.076            | 0.135                 |
| in-samp<br>le<br>(63-<br>contr<br>ast<br>fit) | secre<br>tome         | contrast | 14 | -0.016        | -0.062       | 0.030        | 0.065            | 0.082                 |
| in-samp<br>le<br>(63-<br>contr<br>ast<br>fit) | secre<br>tome         | series   | 8  | -0.026        | -0.107       | 0.055        | 0.112            | 0.082                 |
| leave<br>-<br>class-<br>out                   | repro<br>gram<br>ming | contrast | 7  | 0.021         | -0.140       | 0.182        | 0.222            | 0.135                 |
| leave<br>-<br>class-<br>out                   | repro<br>gram<br>ming | series   | 2  | -0.034        | -1.654       | 1.587        | 2.083            | 0.135                 |

| sche<br>me                                           | class                 | unit     | n  | mean<br>residua<br>l | 95% CI<br>lower | 95%<br>CI<br>upper | detect<br>able<br>shift | class<br>media<br>n  <br>chang<br>e |
|------------------------------------------------------|-----------------------|----------|----|----------------------|-----------------|--------------------|-------------------------|-------------------------------------|
| leave<br>-<br>class-<br>out                          | secre<br>tome         | contrast | 14 | −0.024               | −0.070          | 0.022              | 0.065                   | 0.082                               |
| leave<br>-<br>class-<br>out                          | secre<br>tome         | series   | 8  | −0.034               | −0.116          | 0.048              | 0.113                   | 0.082                               |
| all 21<br>inter<br>venti<br>ons<br>left<br>out       | repro<br>gram<br>ming | contrast | 7  | 0.015                | −0.146          | 0.175              | 0.221                   | 0.135                               |
| all 21<br>inter<br>venti<br>ons<br>left<br>out       | repro<br>gram<br>ming | series   | 2  | −0.040               | −1.655          | 1.575              | 2.076                   | 0.135                               |
| all 21<br>inter<br>venti<br>ons<br>left<br>out       | secre<br>tome         | contrast | 14 | −0.019               | −0.064          | 0.027              | 0.065                   | 0.082                               |
| all 21<br>inter<br>venti<br>ons<br>left<br>out       | secre<br>tome         | series   | 8  | −0.028               | −0.110          | 0.053              | 0.112                   | 0.082                               |
| class<br>and<br>same-<br>serie<br>s<br>contr<br>asts | repro<br>gram<br>ming | contrast | 7  | 0.021                | −0.140          | 0.183              | 0.222                   | 0.135                               |

| scheme                                   | class         | unit     | n  | mean residual | 95% CI lower | 95% CI upper | detectable shift | class median   change |
|------------------------------------------|---------------|----------|----|---------------|--------------|--------------|------------------|-----------------------|
| left out                                 |               |          |    |               |              |              |                  |                       |
| class and same-series contrasts left out | reprogramming | series   | 2  | −0.034        | −1.656       | 1.589        | 2.086            | 0.135                 |
| class and same-series contrasts left out | secretome     | contrast | 14 | −0.024        | −0.070       | 0.022        | 0.065            | 0.082                 |
| class and same-series contrasts left out | secretome     | series   | 8  | −0.034        | −0.116       | 0.048        | 0.113            | 0.082                 |

### Supplementary Table 5. Specificity of the coupling, and the specification range of the residual age effect

All of this table is post hoc.

**(a) The counted libraries.** Correlation with the measured division rate across the 328 counted libraries for each splicing definition and each comparator, the partial correlation of each when the other is held constant, their difference, and the Pearson correlation between the two scores. Every set, the 177-gene set included, is scored in the expression matrix of the

programme audit so that all sets pass one filter; 176 of the 177 genes pass it, and there the 177-gene score correlates with the measured rate at 0.68 (0.71 for the score of Fig. 2a). No comparison shares a gene: shared genes were removed from both sides for the unselected definitions and from the comparator only for the selected set, and where a definition is nested in Metabolism of RNA its genes were removed from that comparator.

| splicing<br>definition | comparator        | genes,<br>splicing | genes,<br>comparator | rho<br>splicing | rho<br>comparator | splicing  <br>comparator | comparator  <br>splicing | asymmetry | %<br>of<br>selected<br>asymmetry | between<br>scores |
|------------------------|-------------------|--------------------|----------------------|-----------------|-------------------|--------------------------|--------------------------|-----------|----------------------------------|-------------------|
| selected_177           | translation       | 176                | 346                  | 0.68            | 0.43              | 0.59                     | -0.10                    | 0.69      | 100                              | 0.68              |
| selected_177           | metabolism_of_rna | 176                | 548                  | 0.68            | 0.47              | 0.58                     | -0.22                    | 0.80      | 100                              | 0.82              |
| selected_177           | rrna_processing   | 176                | 202                  | 0.68            | 0.46              | 0.56                     | -0.05                    | 0.61      | 100                              | 0.69              |
| reactome_mrna_splicing | translation       | 210                | 348                  | 0.63            | 0.43              | 0.51                     | -0.07                    | 0.58      | 85                               | 0.75              |
| reactome_mrna_splicing | metabolism_of_rna | 210                | 514                  | 0.63            | 0.52              | 0.43                     | -0.10                    | 0.53      | 66                               | 0.89              |
| reactome_mrna_splicing | rrna_processing   | 208                | 202                  | 0.63            | 0.46              | 0.48                     | 0.01                     | 0.47      | 78                               | 0.73              |
| reactome_prem_rna_p    | translation       | 283                | 343                  | 0.65            | 0.43              | 0.55                     | -0.08                    | 0.63      | 91                               | 0.70              |

| splicing<br>definition          | comparator        | genes,<br>splicing | genes,<br>comparator | rho<br>splicing | rho<br>comparator | splicing  <br>comparator | comparator  <br>splicing | asymmetry | %<br>of<br>selected<br>asymmetry | rank<br>between<br>scores |
|---------------------------------|-------------------|--------------------|----------------------|-----------------|-------------------|--------------------------|--------------------------|-----------|----------------------------------|---------------------------|
| roces<br>sing                   |                   |                    |                      |                 |                   |                          |                          |           |                                  |                           |
| reactome_prem_rna_processing    | metabolism_of_rna | 288                | 436                  | 0.65            | 0.46              | 0.54                     | -0.18                    | 0.72      | 90                               | 0.83                      |
| reactome_prem_rna_processing    | rrna_processing   | 283                | 199                  | 0.65            | 0.45              | 0.52                     | -0.01                    | 0.52      | 86                               | 0.69                      |
| mrna_splicing_not_selected      | translation       | 84                 | 348                  | 0.31            | 0.43              | 0.06                     | 0.32                     | -0.25     | -37                              | 0.67                      |
| mrna_splicing_not_selected      | metabolism_of_rna | 84                 | 640                  | 0.31            | 0.58              | -0.13                    | 0.53                     | -0.66     | -83                              | 0.71                      |
| mrna_splicing_not_selected      | rrna_processing   | 83                 | 203                  | 0.29            | 0.46              | 0.06                     | 0.37                     | -0.31     | -51                              | 0.60                      |
| premrna_processing_not_selected | translation       | 109                | 345                  | 0.35            | 0.43              | 0.15                     | 0.30                     | -0.15     | -21                              | 0.58                      |

| splicing<br>definition          | comparator        | genes,<br>splicing | genes,<br>comparator | rho<br>splicing | rho<br>comparator | splicing  <br>comparator | comparator  <br>splicing | asymmetry | %<br>of<br>selected<br>asymmetry | between<br>scores |
|---------------------------------|-------------------|--------------------|----------------------|-----------------|-------------------|--------------------------|--------------------------|-----------|----------------------------------|-------------------|
| premrna_processing_not_selected | metabolism_of_rna | 112                | 612                  | 0.34            | 0.58              | -0.07                    | 0.50                     | -0.57     | -72                              | 0.70              |
| premrna_processing_not_selected | rrna_processing   | 109                | 201                  | 0.30            | 0.45              | 0.09                     | 0.37                     | -0.27     | -45                              | 0.53              |

The comparisons affected by the choice of rule, repeated under the other rule: the selected set with shared genes removed from both sides, and the other definitions with shared genes removed from the comparator only.

| splicing<br>definition      | comparator      | shared<br>genes<br>removed from | shared<br>genes | genes,<br>splicing | genes,<br>comparator | splicing  <br>comparator | comparator  <br>splicing | asymmetry |
|-----------------------------|-----------------|---------------------------------|-----------------|--------------------|----------------------|--------------------------|--------------------------|-----------|
| selected_177                | translation     | both sides                      | 2               | 174                | 346                  | 0.587                    | -0.099                   | 0.686     |
| selected_177                | rrna_processing | both sides                      | 2               | 174                | 202                  | 0.561                    | -0.039                   | 0.599     |
| reactome_mrna_splicing      | rrna_processing | comparator only                 | 2               | 210                | 202                  | 0.486                    | 0.010                    | 0.477     |
| reactome_premrna_processing | translation     | comparator only                 | 5               | 288                | 343                  | 0.542                    | -0.078                   | 0.620     |

| splicing<br>definition          | comparator      | shared<br>genes<br>removed from | shared<br>genes | genes,<br>splicing | genes,<br>comparator | splicing  <br>comparator | comparator  <br>splicing | asymmetry |
|---------------------------------|-----------------|---------------------------------|-----------------|--------------------|----------------------|--------------------------|--------------------------|-----------|
| reactome_premrna_processing     | rrna_processing | comparator only                 | 5               | 288                | 199                  | 0.520                    | -0.016                   | 0.536     |
| mrna_splicing_not_selected      | rrna_processing | comparator only                 | 1               | 84                 | 203                  | 0.081                    | 0.362                    | -0.281    |
| premrna_processing_not_selected | translation     | comparator only                 | 3               | 112                | 345                  | 0.114                    | 0.304                    | -0.189    |
| premrna_processing_not_selected | rrna_processing | comparator only                 | 3               | 112                | 201                  | 0.125                    | 0.345                    | -0.219    |

**(b) The donor cohort.** Age effect per decade in the 107 adult donors under each model.

| splicing<br>definition | comparator        | side       | metric            | model                     | genes | per decade | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|------------------------|-------------------|------------|-------------------|---------------------------|-------|------------|--------------------|--------------------|---------|
| selected_177           |                   | splicing   | splicing          | technical covariates only | 177   | -0.133     | -0.183             | -0.084             | 5.9e-07 |
| selected_177           | translation       | comparator | translation       | technical covariates only | 363   | -0.067     | -0.100             | -0.034             | 0.00011 |
| selected_177           | metabolism_of_rna | comparator | metabolism_of_rna | technical covariates only | 576   | -0.056     | -0.081             | -0.030             | 3.3e-05 |

| splicing<br>definition | comparative       | side       | metric            | model                         | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|------------------------|-------------------|------------|-------------------|-------------------------------|-------|------------|--------------------|--------------------|---------|
| selected_177           | rrna_processing   | comparator | rrna_processing   | technical covariates only     | 229   | -0.074     | -0.111             | -0.037             | 0.00013 |
| selected_177           |                   | splicing   | splicing          | + proliferation score         | 177   | -0.040     | -0.068             | -0.013             | 0.0049  |
| selected_177           | translation       | comparator | translation       | + proliferation score         | 363   | -0.032     | -0.063             | 0.000              | 0.05    |
| selected_177           | metabolism_of_rna | comparator | metabolism_of_rna | + proliferation score         | 576   | -0.024     | -0.047             | -0.001             | 0.039   |
| selected_177           | rrna_processing   | comparator | rrna_processing   | + proliferation score         | 229   | -0.030     | -0.064             | 0.004              | 0.079   |
| selected_177           | translation       | splicing   | splicing          | + proliferation + translation | 177   | -0.022     | -0.043             | -0.001             | 0.044   |
| selected_177           | translation       | comparator | translation       | + proliferation + splicing    | 363   | -0.001     | -0.025             | 0.024              | 0.94    |
| selected_177           | metabolism_of_rna | splicing   | splicing          | + proliferation               | 177   | -0.016     | -0.033             | -0.000             | 0.05    |

| splicing<br>definition         | comparator            | side       | metric                | model                                      | genes   | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P           |
|--------------------------------|-----------------------|------------|-----------------------|--------------------------------------------|---------|------------|--------------------|--------------------|-------------|
|                                | rna                   |            |                       | n +<br>Metabolism of<br>RNA                |         |            |                    |                    |             |
| select<br>ed_17<br>7           | metabolism_of_<br>rna | comparator | metabolism_of_<br>rna | +<br>proliferation +<br>splicing           | 57<br>6 | 0.00<br>3  | −0.0<br>11         | 0.01<br>7          | 0.66        |
| select<br>ed_17<br>7           | rrna_<br>processing   | splicing   | splicing              | +<br>proliferation +<br>rRNA<br>processing | 17<br>7 | −0.0<br>22 | −0.0<br>42         | −0.0<br>03         | 0.026       |
| select<br>ed_17<br>7           | rrna_<br>processing   | comparator | rrna_<br>processing   | +<br>proliferation +<br>splicing           | 22<br>9 | 0.00<br>5  | −0.0<br>19         | 0.03<br>0          | 0.68        |
| reactome_<br>mrna_<br>splicing |                       | splicing   | splicing              | technical<br>covariates<br>only            | 20<br>9 | −0.0<br>94 | −0.1<br>28         | −0.0<br>60         | 2.5e-<br>07 |
| reactome_<br>mrna_<br>splicing | translation           | comparator | translation           | technical<br>covariates<br>only            | 36<br>5 | −0.0<br>68 | −0.1<br>01         | −0.0<br>34         | 0.000<br>11 |
| reactome_<br>mrna_<br>splicing | metabolism_of_<br>rna | comparator | metabolism_of_<br>rna | technical<br>covariates<br>only            | 54<br>4 | −0.0<br>66 | −0.0<br>96         | −0.0<br>37         | 1.7e-<br>05 |
| react                          | rrna                  | comparator | rrna                  | techn                                      | 22      | −0.0       | −0.1               | −0.0               | 0.000       |

| splicing<br>definition  | comparator        | side       | metric            | model                         | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P      |
|-------------------------|-------------------|------------|-------------------|-------------------------------|-------|------------|--------------------|--------------------|--------|
| ome_mrna_splicing       | processing        | or         | processing        | ical covariates only          | 9     | 75         | 12                 | 38                 | 13     |
| react_ome_mrna_splicing |                   | splicing   | splicing          | + proliferation score         | 209   | -0.042     | -0.068             | -0.016             | 0.0019 |
| react_ome_mrna_splicing | translation       | comparator | translation       | + proliferation score         | 365   | -0.032     | -0.063             | -0.000             | 0.05   |
| react_ome_mrna_splicing | metabolism_of_rna | comparator | metabolism_of_rna | + proliferation score         | 544   | -0.023     | -0.046             | 0.001              | 0.056  |
| react_ome_mrna_splicing | rrna_processing   | comparator | rrna_processing   | + proliferation score         | 229   | -0.030     | -0.064             | 0.004              | 0.08   |
| react_ome_mrna_splicing | translation       | splicing   | splicing          | + proliferation + translation | 209   | -0.023     | -0.042             | -0.005             | 0.015  |
| react_ome_mrna_splicing | translation       | comparator | translation       | + proliferation + splicing    | 365   | 0.004      | -0.019             | 0.028              | 0.71   |
| react_ome_mrna_splicing | metabolism_of_rna | splicing   | splicing          | + proliferation +             | 209   | -0.020     | -0.034             | -0.006             | 0.0051 |

| splicing<br>definition      | comparative       | side       | metric            | model                             | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|-----------------------------|-------------------|------------|-------------------|-----------------------------------|-------|------------|--------------------|--------------------|---------|
| ing                         | rna               |            |                   | Metabolism of RNA                 |       |            |                    |                    |         |
| reactome_mrna_splicing      | metabolism_of_rna | comparator | metabolism_of_rna | + proliferation + splicing        | 544   | 0.009      | -0.004             | 0.022              | 0.17    |
| reactome_mrna_splicing      | rrna_processing   | splicing   | splicing          | + proliferation + rRNA processing | 207   | -0.025     | -0.044             | -0.007             | 0.0075  |
| reactome_mrna_splicing      | rrna_processing   | comparator | rrna_processing   | + proliferation + splicing        | 229   | 0.009      | -0.015             | 0.034              | 0.45    |
| reactome_premrna_processing |                   | splicing   | splicing          | technical covariates only         | 286   | -0.096     | -0.131             | -0.061             | 4e-07   |
| reactome_premrna_processing | translation       | comparator | translation       | technical covariates only         | 360   | -0.068     | -0.101             | -0.035             | 0.0001  |
| reactome_premrna_processing | metabolism_of_rna | comparator | metabolism_of_rna | technical covariates only         | 467   | -0.061     | -0.089             | -0.032             | 4.8e-05 |

| splicing<br>definition      | comparative<br>side | metric          | model                         | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|-----------------------------|---------------------|-----------------|-------------------------------|-------|------------|--------------------|--------------------|---------|
| single                      |                     |                 |                               |       |            |                    |                    |         |
| reactome_premrna_processing | comparator          | rrna_processing | technical covariates only     | 226   | -0.075     | -0.112             | -0.038             | 0.00012 |
| reactome_premrna_processing | splicing            | splicing        | + proliferation score         | 286   | -0.037     | -0.062             | -0.013             | 0.0035  |
| reactome_premrna_processing | translation         | comparator      | + proliferation score         | 360   | -0.032     | -0.063             | -0.000             | 0.05    |
| reactome_premrna_processing | metabolism_of_rna   | comparator      | + proliferation score         | 467   | -0.022     | -0.047             | 0.002              | 0.073   |
| reactome_premrna_processing | rrna_processing     | comparator      | + proliferation score         | 226   | -0.030     | -0.064             | 0.004              | 0.08    |
| reactome_premrna_processing | translation         | splicing        | + proliferation + translation | 281   | -0.021     | -0.041             | -0.002             | 0.032   |
| reactome_premrna_processing | translation         | comparator      | + proliferation               | 360   | -0.001     | -0.027             | 0.024              | 0.91    |

| splicing<br>definition       | comparative       | side       | metric            | model                                        | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|------------------------------|-------------------|------------|-------------------|----------------------------------------------|-------|------------|--------------------|--------------------|---------|
| rna_processing               |                   |            |                   | n +<br>splicing                              |       |            |                    |                    |         |
| reactome_prem_rna_processing | metabolism_of_rna | splicing   | splicing          | +<br>proliferation +<br>Metabolism of<br>RNA | 286   | -0.020     | -0.035             | -0.004             | 0.015   |
| reactome_prem_rna_processing | metabolism_of_rna | comparator | metabolism_of_rna | +<br>proliferation +<br>splicing             | 467   | 0.007      | -0.009             | 0.022              | 0.42    |
| reactome_prem_rna_processing | rrna_processing   | splicing   | splicing          | +<br>proliferation +<br>rRNA<br>processing   | 281   | -0.023     | -0.041             | -0.004             | 0.018   |
| reactome_prem_rna_processing | rrna_processing   | comparator | rrna_processing   | +<br>proliferation +<br>splicing             | 226   | 0.004      | -0.022             | 0.031              | 0.75    |
| mrna_splicing_not_selected   |                   | splicing   | splicing          | technical<br>covariates<br>only              | 82    | -0.035     | -0.060             | -0.010             | 0.0059  |
| mrna_splicing_not_selected   | translation       | comparator | translation       | technical<br>covariates<br>only              | 365   | -0.068     | -0.101             | -0.034             | 0.00011 |

| splicing<br>definition     | comparator        | side       | metric            | model                         | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|----------------------------|-------------------|------------|-------------------|-------------------------------|-------|------------|--------------------|--------------------|---------|
| mrna_splicing_not_selected | metabolism_of_rna | comparator | metabolism_of_rna | technical covariates only     | 671   | -0.079     | -0.111             | -0.047             | 4.5e-06 |
| mrna_splicing_not_selected | rrna_processing   | comparator | rrna_processing   | technical covariates only     | 230   | -0.075     | -0.112             | -0.037             | 0.00013 |
| mrna_splicing_not_selected |                   | splicing   | splicing          | + proliferation score         | 82    | -0.038     | -0.065             | -0.011             | 0.0063  |
| mrna_splicing_not_selected | translation       | comparator | translation       | + proliferation score         | 365   | -0.032     | -0.063             | -0.000             | 0.05    |
| mrna_splicing_not_selected | metabolism_of_rna | comparator | metabolism_of_rna | + proliferation score         | 671   | -0.027     | -0.050             | -0.003             | 0.028   |
| mrna_splicing_not_selected | rrna_processing   | comparator | rrna_processing   | + proliferation score         | 230   | -0.030     | -0.064             | 0.004              | 0.083   |
| mrna_splicing_not_selected | translation       | splicing   | splicing          | + proliferation + translation | 82    | -0.024     | -0.048             | -0.000             | 0.047   |
| mrna_splicing_not_selected | translation       | comparator | translation       | + proliferation +             | 365   | -0.008     | -0.037             | 0.020              | 0.55    |

| splicing<br>definition           | comparato<br>r    | side       | metric            | model                               | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|----------------------------------|-------------------|------------|-------------------|-------------------------------------|-------|------------|--------------------|--------------------|---------|
| ected                            |                   |            |                   | splicing                            |       |            |                    |                    |         |
| mrna_splicing_not_selected       | metabolism_of_rna | splicing   | splicing          | + proliferation + Metabolism of RNA | 82    | -0.018     | -0.040             | 0.003              | 0.089   |
| mrna_splicing_not_selected       | metabolism_of_rna | comparator | metabolism_of_rna | + proliferation + splicing          | 671   | -0.005     | -0.024             | 0.014              | 0.6     |
| mrna_splicing_not_selected       | rrna_processing   | splicing   | splicing          | + proliferation + rRNA processing   | 81    | -0.026     | -0.050             | -0.002             | 0.035   |
| mrna_splicing_not_selected       | rrna_processing   | comparator | rrna_processing   | + proliferation + splicing          | 230   | -0.007     | -0.038             | 0.024              | 0.67    |
| prem_rna_processing_not_selected |                   | splicing   | splicing          | technical covariates only           | 109   | -0.034     | -0.057             | -0.011             | 0.0043  |
| prem_rna_processing_             | translation       | comparator | translation       | technical covariates                | 362   | -0.068     | -0.102             | -0.035             | 9.8e-05 |

| splicing<br>definition          | comparator        | side       | metric            | model                     | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P       |
|---------------------------------|-------------------|------------|-------------------|---------------------------|-------|------------|--------------------|--------------------|---------|
| not_selected                    |                   |            |                   | only                      |       |            |                    |                    |         |
| prem_rna_processes_not_selected | metabolism_of_rna | comparator | metabolism_of_rna | technical_covariates_only | 644   | −0.081     | −0.114             | −0.048             | 4.3e-06 |
| prem_rna_processes_not_selected | rrna_processing   | comparator | rrna_processing   | technical_covariates_only | 228   | −0.075     | −0.112             | −0.038             | 0.00012 |
| prem_rna_processes_not_selected |                   | splicing   | splicing          | + proliferation score     | 109   | −0.032     | −0.058             | −0.007             | 0.013   |
| prem_rna_processes_not_selected | translation       | comparator | translation       | + proliferation score     | 362   | −0.032     | −0.063             | −0.000             | 0.05    |
| prem_rna_processes_not_selected | metabolism_of_rna | comparator | metabolism_of_rna | + proliferation score     | 644   | −0.027     | −0.051             | −0.003             | 0.028   |
| prem_rna p                      | rrna_proce        | comparator | rrna_proce        | + prolif                  | 228   | −0.030     | −0.065             | 0.004              | 0.08    |

| splicing<br>definition                             | comparative       | side       | metric            | model                               | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P     |
|----------------------------------------------------|-------------------|------------|-------------------|-------------------------------------|-------|------------|--------------------|--------------------|-------|
| proteins<br>sing_<br>not_<br>selected              | ssing             |            | ssing             | eration<br>score                    |       |            |                    |                    |       |
| premature<br>proteins<br>sing_<br>not_<br>selected | translation       | splicing   | splicing          | + proliferation + translation       | 106   | -0.021     | -0.045             | 0.003              | 0.084 |
| premature<br>proteins<br>sing_<br>not_<br>selected | translation       | comparator | translation       | + proliferation + splicing          | 362   | -0.014     | -0.043             | 0.016              | 0.35  |
| premature<br>proteins<br>sing_<br>not_<br>selected | metabolism_of_rna | splicing   | splicing          | + proliferation + Metabolism of RNA | 109   | -0.014     | -0.034             | 0.006              | 0.17  |
| premature<br>proteins<br>sing_<br>not_<br>selected | metabolism_of_rna | comparator | metabolism_of_rna | + proliferation + splicing          | 644   | -0.007     | -0.026             | 0.012              | 0.46  |
| premature<br>proteins<br>sing_<br>not_<br>selected | rRNA_processing   | splicing   | splicing          | + proliferation + rRNA processing   | 106   | -0.022     | -0.045             | 0.001              | 0.057 |

| splicing<br>definition           | comparator      | side       | metric          | model                      | genes | percentage | 95%<br>CI<br>lower | 95%<br>CI<br>upper | P    |
|----------------------------------|-----------------|------------|-----------------|----------------------------|-------|------------|--------------------|--------------------|------|
| selected                         |                 |            |                 | splicing                   |       |            |                    |                    |      |
| prem_rna_processing_not_selected | rrna_processing | comparator | rrna_processing | + proliferation + splicing | 228   | -0.10      | -0.42              | 0.021              | 0.52 |

**(c) GTEx cultures.** The same comparison against the 20-marker transcriptional proliferation score in 511 cultures. The asymmetry is not reproduced.

| splicing<br>definition       | comparator        | n   | rho<br>splicing | rho<br>comparator | splicing  <br>comparator | comparator<br> <br>splicing | asymmetry |
|------------------------------|-------------------|-----|-----------------|-------------------|--------------------------|-----------------------------|-----------|
| selected_177                 | translation       | 511 | 0.75            | 0.78              | 0.41                     | 0.51                        | -0.10     |
| selected_177                 | metabolism_of_rna | 511 | 0.75            | 0.62              | 0.60                     | -0.29                       | 0.88      |
| selected_177                 | rrna_processing   | 511 | 0.75            | 0.66              | 0.53                     | 0.26                        | 0.27      |
| reactome_mrna_splicing       | translation       | 511 | 0.68            | 0.78              | 0.28                     | 0.57                        | -0.29     |
| reactome_mrna_splicing       | metabolism_of_rna | 511 | 0.68            | 0.66              | 0.25                     | 0.10                        | 0.14      |
| reactome_mrna_splicing       | rrna_processing   | 511 | 0.68            | 0.66              | 0.41                     | 0.35                        | 0.06      |
| reactome_pre_mrna_processing | translation       | 511 | 0.68            | 0.78              | 0.29                     | 0.59                        | -0.30     |
| reactome_pre_mrna_processing | metabolism_of_rna | 511 | 0.68            | 0.66              | 0.28                     | 0.11                        | 0.16      |
| reactome_pre_mrna_processing | rrna_processing   | 511 | 0.68            | 0.65              | 0.40                     | 0.34                        | 0.06      |
| mrna_splicing                | translation       | 511 | 0.43            | 0.78              | 0.00                     | 0.72                        | -0.72     |

| splicing<br>definition                  | comparator            | n   | rho<br>splici<br>ng | rho<br>com<br>para<br>tor | splici<br>ng  <br>comp<br>arator | comp<br>arator<br> <br>splici<br>ng | asym<br>metry |
|-----------------------------------------|-----------------------|-----|---------------------|---------------------------|----------------------------------|-------------------------------------|---------------|
| _not_selected                           |                       |     |                     |                           |                                  |                                     |               |
| mrna_splicing<br>_not_selected          | metabolism_o<br>f_rna | 511 | 0.43                | 0.70                      | -0.44                            | 0.70                                | -1.14         |
| mrna_splicing<br>_not_selected          | rrna_processi<br>ng   | 511 | 0.42                | 0.66                      | 0.07                             | 0.57                                | -0.50         |
| premrna_proc<br>essing_not_se<br>lected | translation           | 511 | 0.40                | 0.78                      | 0.01                             | 0.73                                | -0.73         |
| premrna_proc<br>essing_not_se<br>lected | metabolism_o<br>f_rna | 511 | 0.41                | 0.71                      | -0.54                            | 0.76                                | -1.31         |
| premrna_proc<br>essing_not_se<br>lected | rrna_processi<br>ng   | 511 | 0.38                | 0.65                      | 0.01                             | 0.58                                | -0.56         |

**(d) The two cohorts, not pooled.** Proliferation-adjusted age effects of the two fibroblast cohorts, reported side by side; they are not combined, because one is an attenuation and the other a suppression effect in cultures with no unadjusted age effect.

| gene set                           | cohort                            | n   | per<br>decade | 95% CI<br>lower | 95% CI<br>upper | P      |
|------------------------------------|-----------------------------------|-----|---------------|-----------------|-----------------|--------|
| preregistere<br>d 96-gene set      | GSE113957<br>fibroblast<br>donors | 107 | -0.0330       | -0.0644         | -0.0016         | 0.04   |
| preregistere<br>d 96-gene set      | GTEx<br>cultured<br>fibroblasts   | 492 | -0.0529       | -0.0866         | -0.0192         | 0.0022 |
| Reactome<br>pre-mRNA<br>processing | GSE113957<br>fibroblast<br>donors | 107 | -0.0373       | -0.0620         | -0.0126         | 0.0035 |
| Reactome<br>pre-mRNA<br>processing | GTEx<br>cultured<br>fibroblasts   | 492 | -0.0448       | -0.0729         | -0.0167         | 0.0018 |

**(e) The four definitions of the fibroblast cohort,** reported as sensitivity and not pooled, since they are not independent of one another.

| metric      | cohort    | n   | adjusted<br>per<br>decade | 95% CI<br>lower | 95% CI<br>upper | P       |
|-------------|-----------|-----|---------------------------|-----------------|-----------------|---------|
| machinery   | adult2082 | 76  | -0.056                    | -0.095          | -0.017          | 0.0057  |
| machinery   | normal    | 133 | -0.045                    | -0.069          | -0.020          | 0.00044 |
| machinery   | primary   | 107 | -0.040                    | -0.068          | -0.013          | 0.0049  |
| machinery   | published | 142 | -0.023                    | -0.044          | -0.002          | 0.035   |
| machinery96 | adult2082 | 76  | -0.068                    | -0.115          | -0.021          | 0.0053  |
| machinery96 | normal    | 133 | -0.058                    | -0.085          | -0.032          | 3.3e-05 |
| machinery96 | primary   | 107 | -0.033                    | -0.064          | -0.002          | 0.04    |
| machinery96 | published | 142 | -0.031                    | -0.053          | -0.009          | 0.0069  |

**(f) Every sample-level step in one unit (post hoc).** The 20-marker proliferation score calibrated against the counted division rate in the counted libraries, so that each dataset's slope can be read in divisions per day. Score-based slopes are multiplied by 0.51, the factor by which the naive conversion overstates the effect where counting is available.

| dataset                             | n   | per SD of<br>proliferation | per<br>div.<br>per<br>day | rho   |
|-------------------------------------|-----|----------------------------|---------------------------|-------|
| counted resource, against counting  | 328 | 0.424                      | 2.90                      | 0.705 |
| counted resource, against the score | 328 | 0.830                      | 2.90                      | 0.842 |
| donor cohort                        | 107 | 0.891                      | 3.11                      | 0.912 |
| GTEEx cultured fibroblasts          | 492 | 0.738                      | 2.58                      | 0.739 |
| GTEEx skin, sun-exposed             | 739 | 0.247                      | 0.86                      | 0.225 |
| GTEEx skin, not exposed             | 626 | 0.108                      | 0.38                      | 0.056 |

### Supplementary Table 6. Ageing programmes against measured division rate and donor age

First: correlation with the measured division rate across 328 counted libraries, with all genes and after removing every gene belonging to the Reactome cell-cycle union. Second: the same programmes against donor age in the 107-donor primary cohort, before and after adjustment for proliferation (adj), each with its nominal P value and BH-FDR.

| programme                         | genes | cell-cycle<br>genes<br>removed | rho,<br>all<br>genes | rho,<br>after<br>removal | FDR     |
|-----------------------------------|-------|--------------------------------|----------------------|--------------------------|---------|
| chromatin<br>organisation         | 312   | 38                             | 0.69                 | 0.63                     | 1.3e-36 |
| cellular senescence               | 150   | 82                             | 0.67                 | 0.38                     | 1.6e-12 |
| pre-mRNA processing               | 287   | 46                             | 0.65                 | 0.67                     | 5.9e-42 |
| mRNA splicing                     | 198   | 12                             | 0.63                 | 0.64                     | 8.5e-39 |
| cell cycle (positive<br>control)  | 499   | 499                            | 0.62                 |                          |         |
| interferon signalling             | 279   | 67                             | 0.60                 | 0.49                     | 1.7e-20 |
| DNA double-strand<br>repair       | 180   | 124                            | 0.59                 | 0.55                     | 1.7e-26 |
| telomere maintenance              | 82    | 82                             | 0.58                 |                          |         |
| base excision repair              | 63    | 44                             | 0.56                 | 0.35                     | 2.1e-10 |
| interleukin signalling            | 334   | 65                             | 0.47                 | 0.26                     | 2.8e-06 |
| rRNA processing                   | 187   | 9                              | 0.46                 | 0.45                     | 5.5e-17 |
| mitochondrial<br>translation      | 100   | 0                              | 0.45                 | 0.45                     | 1.4e-17 |
| translation                       | 348   | 39                             | 0.43                 | 0.42                     | 1.2e-14 |
| extracellular matrix              | 235   | 1                              | -0.33                | -0.34                    | 7.4e-10 |
| TCA cycle                         | 34    | 0                              | 0.29                 | 0.29                     | 9.3e-08 |
| insulin / IGF signalling          | 62    | 3                              | -0.27                | -0.31                    | 2e-08   |
| respiratory electron<br>transport | 142   | 0                              | 0.22                 | 0.22                     | 5.7e-05 |
| proteasome                        | 268   | 64                             | 0.22                 | -0.08                    | 0.16    |
| chaperone / HSF1                  | 30    | 5                              | 0.21                 | 0.10                     | 0.073   |
| NF-kB / TNF                       | 32    | 5                              | -0.14                | -0.23                    | 3.9e-05 |
| autophagy                         | 130   | 38                             | 0.12                 | -0.03                    | 0.65    |
| mTOR signalling                   | 23    | 0                              | 0.11                 | 0.11                     | 0.062   |
| unfolded protein<br>response      | 89    | 3                              | 0.05                 | 0.05                     | 0.4     |
| collagen formation                | 71    | 0                              | -0.02                | -0.02                    | 0.76    |

| program<br>me | rho<br>with<br>age | per<br>decade | P       | FDR     | per<br>decade<br>(adj) | P (adj) | FDR<br>(adj) |
|---------------|--------------------|---------------|---------|---------|------------------------|---------|--------------|
| chromati      | -0.51              | -0.038        | 1.7e-05 | 8.3e-05 | -0.015                 | 0.037   | 0.25         |

| program<br>me                        | rho<br>with<br>age | per<br>decade | P       | FDR     | per<br>decade<br>(adj) | P (adj) | FDR<br>(adj) |
|--------------------------------------|--------------------|---------------|---------|---------|------------------------|---------|--------------|
| n<br>organisat<br>ion                |                    |               |         |         |                        |         |              |
| cellular<br>senescen<br>ce           | −0.37              | −0.026        | 0.0024  | 0.0044  | 0.000                  | 1       | 1            |
| pre-<br>mRNA<br>processin<br>g       | −0.49              | −0.096        | 3.8e-07 | 4.6e-06 | −0.037                 | 0.0034  | 0.04         |
| mRNA<br>splicing                     | −0.51              | −0.098        | 2.1e-07 | 4.6e-06 | −0.043                 | 0.0017  | 0.04         |
| cell cycle<br>(positive<br>control)  | −0.47              | −0.088        | 2.9e-05 | 0.0001  | −0.006                 | 0.42    | 0.64         |
| interfero<br>n<br>signalling         | −0.30              | −0.027        | 0.0024  | 0.0044  | −0.006                 | 0.46    | 0.64         |
| DNA<br>double-<br>strand<br>repair   | −0.47              | −0.099        | 9.1e-06 | 7.3e-05 | −0.014                 | 0.11    | 0.35         |
| telomere<br>maintena<br>nce          | −0.49              | −0.096        | 1.4e-05 | 8.3e-05 | −0.011                 | 0.19    | 0.38         |
| base<br>excision<br>repair           | −0.50              | −0.097        | 2.3e-05 | 9.1e-05 | −0.009                 | 0.3     | 0.51         |
| interleuki<br>n<br>signalling        | −0.19              | −0.006        | 0.3     | 0.33    | 0.002                  | 0.78    | 0.94         |
| rRNA<br>processin<br>g               | −0.42              | −0.077        | 0.00017 | 0.0004  | −0.038                 | 0.052   | 0.25         |
| mitochon<br>drial<br>translatio<br>n | −0.42              | −0.082        | 9.5e-05 | 0.00028 | −0.022                 | 0.18    | 0.38         |
| translatio                           | −0.46              | −0.068        | 0.00011 | 0.00029 | −0.032                 | 0.05    | 0.25         |

| program<br>me                         | rho<br>with<br>age | per<br>decade | P      | FDR    | per<br>decade<br>(adj) | P (adj) | FDR<br>(adj) |
|---------------------------------------|--------------------|---------------|--------|--------|------------------------|---------|--------------|
| n                                     |                    |               |        |        |                        |         |              |
| TCA cycle                             | −0.18              | −0.020        | 0.19   | 0.25   | 0.007                  | 0.66    | 0.88         |
| respirator<br>y electron<br>transport | −0.28              | −0.018        | 0.19   | 0.25   | −0.001                 | 0.93    | 0.98         |
| proteaso<br>me                        | −0.01              | 0.011         | 0.093  | 0.14   | 0.009                  | 0.22    | 0.4          |
| chaperon<br>e / HSF1                  | −0.23              | −0.008        | 0.55   | 0.57   | 0.011                  | 0.38    | 0.61         |
| autophag<br>y                         | −0.07              | −0.001        | 0.95   | 0.95   | −0.003                 | 0.78    | 0.94         |
| mTOR<br>signalling                    | −0.14              | −0.012        | 0.31   | 0.33   | 0.001                  | 0.94    | 0.98         |
| unfolded<br>protein<br>response       | −0.43              | −0.020        | 0.044  | 0.07   | −0.016                 | 0.13    | 0.35         |
| collagen<br>formation                 | −0.44              | −0.015        | 0.21   | 0.25   | −0.020                 | 0.13    | 0.35         |
| NF-kB /<br>TNF                        | 0.13               | 0.028         | 0.0023 | 0.0044 | 0.014                  | 0.12    | 0.35         |
| insulin /<br>IGF<br>signalling        | 0.08               | 0.020         | 0.025  | 0.044  | 0.012                  | 0.18    | 0.38         |
| extracellu<br>lar matrix              | −0.18              | 0.013         | 0.15   | 0.21   | −0.002                 | 0.85    | 0.98         |

### Supplementary Table 7. Senescence panels in cultured fibroblasts

Post hoc. Cell-cycle content is the share of a panel's measured genes belonging to the Reactome cell-cycle union. Correlations are with the measured division rate across 328 counted libraries; age effects are per decade in the 107 adult donors. The percentage of the age effect lost is meaningful only for the five panels that had one.

| panel                        | measured | % cell cycle | rho rate | rho after removal | median gene rho | age per decade | FDR     | adjusted | FD R adj | % of age effect lost |
|------------------------------|----------|--------------|----------|-------------------|-----------------|----------------|---------|----------|----------|----------------------|
| Reactome Cellular Senescence | 150      | 55           | 0.67     | 0.38              | 0.21            | -0.026         | 0.0041  | 0.000    | 1        | 100                  |
| SenMayo                      | 88       | 1            | -0.10    | -0.14             | -0.05           | 0.003          | 0.72    | 0.002    | 1        | -                    |
| Fridman senescence, up       | 71       | 15           | -0.07    | 0.01              | -0.07           | 0.043          | 0.00039 | 0.015    | 0.93     | 66                   |
| Fridman senescence, down     | 11       | 27           | 0.38     | 0.01              | 0.16            | -0.034         | 0.02    | -0.005   | 1        | 87                   |
| CellAge, induces senescence  | 315      | 10           | 0.33     | 0.21              | 0.02            | 0.002          | 0.72    | 0.004    | 1        | -                    |
| CellAge, inhibits senescence | 440      | 21           | 0.69     | 0.71              | 0.20            | -0.041         | 0.00037 | -0.002   | 1        | 95                   |
| Reactome Cell Cycle          | 499      | 100          | 0.62     |                   | 0.34            | -0.088         | 0.00021 | -0.006   | 1        | 93                   |

| panel               | me<br>asu<br>red | %<br>cell<br>cyc<br>le | rho<br>rate | rho<br>after<br>remo<br>val | medi<br>an<br>gene<br>rho | age<br>per<br>decad<br>e | FDR | adjust<br>ed | FD<br>R<br>adj | %<br>of<br>ag<br>e<br>eff<br>ect<br>lost |
|---------------------|------------------|------------------------|-------------|-----------------------------|---------------------------|--------------------------|-----|--------------|----------------|------------------------------------------|
| , Mitotic (control) |                  |                        |             |                             |                           |                          |     |              |                |                                          |

### Supplementary Table 8. Preregistered methylation clock tests

The four hypotheses fixed in Supplementary Note 2, each with the criterion it was given. M1 and M2 are Spearman correlations across methylation samples; M3 and M4 are across the 109 cultures carrying both a clock and transcriptional scores. “Met” is the preregistered criterion, which for M1 to M3 is  $|\text{rho}| \geq 0.3$  with a significant P value and for M4 is an attenuation of at least 50% when the measured division rate is held constant.

| test                                                    | rho    | P       | FDR    | n   | met       |
|---------------------------------------------------------|--------|---------|--------|-----|-----------|
| M1 DNAm age vs cumulative doublings (untreated healthy) | 0.356  | 0.00077 | 0.0015 | 86  | TRUE      |
| M2 clock acceleration vs measured division rate         | −0.014 | 0.76    | 0.76   | 471 | FALS<br>E |
| M3 proliferation score vs DNAm age (paired cultures)    | 0.162  | 0.093   | 0.12   | 109 | FALS<br>E |
| M4 splicing vs DNAm age, division rate held constant    | −0.338 | 0.00032 | 0.0013 | 109 | FALS<br>E |

### Supplementary Table 9. The preregistered GTEx hypotheses and their outcomes

The six hypotheses as they were locked before the data were downloaded, with the criterion fixed in advance, the value observed and the verdict. Four were supported; H4 could not be evaluated because the GTEx cultures had no unadjusted age effect to attenuate; and H6 — a positive control asking whether collagen formation falls with age in sun-exposed skin — was not supported, which under the preregistered rule bars age effects measured within GTEx tissues from supporting any claim in this paper. The full

document, its English translation and its ten recorded deviations are Supplementary Note 1.

| hypothesis | test                                                                 | criterion              | observed                                 | verdict                                           |
|------------|----------------------------------------------------------------------|------------------------|------------------------------------------|---------------------------------------------------|
| H1         | culture:<br>splicing genes<br>track<br>proliferation                 | $\rho \geq 0.5$        | 0.74                                     | supported                                         |
| H2         | skin: coupling<br>weaker than in<br>culture                          | one-sided P<br>< 0.05  | 0.23 vs 0.74                             | supported                                         |
| H3         | skeletal<br>muscle: no<br>coupling                                   | $\rho < 0.3$           | -0.12                                    | supported                                         |
| H4         | culture: age<br>effect shrinks<br>with<br>adjustment                 | $\geq 50\%$<br>smaller | no<br>unadjusted<br>effect (P =<br>0.33) | not evaluable<br>(no<br>unadjusted<br>age effect) |
| H5         | culture:<br>collagen<br>formation<br>independent of<br>proliferation | $ \rho  < 0.3$         | 0.05                                     | supported                                         |
| H6         | skin: collagen<br>formation falls<br>with age                        | negative, P<br>< 0.05  | +0.027 per<br>decade (P =<br>0.018)      | not<br>supported                                  |

### Supplementary Table 10. Programmes against proliferation and donor age in GTEx

**(a) Partial correlation with proliferation.** Partial Spearman correlation of each of the twenty-five programme scores with the proliferation score in each tissue, after regressing out RNA integrity, ischaemic time and library size (the values of Fig. 7b).

| programme              | cultured<br>fibroblasts | skin, sun-<br>exposed | skin, not<br>exposed | skeletal<br>muscle |
|------------------------|-------------------------|-----------------------|----------------------|--------------------|
| splicing 96            | 0.74                    | 0.23                  | 0.06                 | -0.12              |
| pre-mRNA<br>processing | 0.69                    | 0.14                  | -0.02                | -0.13              |
| mRNA splicing          | 0.69                    | 0.13                  | -0.04                | -0.14              |
| translation            | 0.75                    | 0.21                  | 0.09                 | -0.21              |
| rRNA processing        | 0.70                    | 0.18                  | 0.01                 | -0.11              |

| programme                      | cultured fibroblasts | skin, sun-exposed | skin, not exposed | skeletal muscle |
|--------------------------------|----------------------|-------------------|-------------------|-----------------|
| proteasome                     | 0.27                 | 0.16              | 0.02              | −0.37           |
| autophagy                      | 0.44                 | 0.06              | −0.10             | −0.33           |
| chaperone / HSF1               | 0.17                 | 0.21              | 0.07              | −0.31           |
| unfolded protein response      | 0.70                 | 0.03              | −0.08             | −0.16           |
| DNA double-strand repair       | 0.90                 | 0.64              | 0.58              | −0.05           |
| base excision repair           | 0.85                 | 0.62              | 0.58              | 0.30            |
| telomere maintenance           | 0.81                 | 0.61              | 0.52              | 0.09            |
| chromatin organisation         | 0.43                 | 0.14              | 0.02              | −0.31           |
| mitochondrial translation      | 0.42                 | 0.23              | 0.13              | −0.15           |
| respiratory electron transport | 0.67                 | 0.26              | 0.24              | −0.12           |
| TCA cycle                      | 0.74                 | 0.19              | 0.18              | −0.10           |
| interferon signalling          | 0.59                 | 0.24              | 0.06              | 0.01            |
| NF-κB / TNF                    | −0.56                | 0.04              | −0.12             | −0.36           |
| extracellular matrix           | −0.21                | −0.10             | −0.01             | 0.22            |
| collagen formation             | 0.05                 | 0.03              | 0.09              | 0.24            |
| insulin / IGF signalling       | −0.08                | −0.11             | −0.11             | −0.07           |
| mTOR signalling                | 0.41                 | 0.13              | −0.01             | −0.31           |
| cellular senescence            | 0.63                 | 0.19              | 0.05              | −0.21           |
| interleukin signalling         | 0.42                 | −0.02             | −0.09             | −0.13           |
| cell cycle (positive control)  | 0.94                 | 0.67              | 0.60              | 0.03            |

**(b) Age effects before and after adjustment for proliferation.** Change in each programme score per decade of donor age, from linear models with sex, Hardy death classification, RNA integrity, ischaemic time and library size as covariates, and the same with the proliferation score added (adj). FDR is Benjamini-Hochberg within a tissue and was computed for the unadjusted effects. The preregistered positive control for age effects in tissue failed (H6), so under the preregistered rule the age effects in skin and muscle are listed for completeness and support no claim.

*Cultured fibroblasts* (492 donors).

| programme                         | per<br>decade | P       | FDR  | per<br>decade<br>(adj) | P (adj) |
|-----------------------------------|---------------|---------|------|------------------------|---------|
| splicing 96                       | −0.025        | 0.33    | 0.51 | −0.053                 | 0.0022  |
| pre-mRNA processing               | −0.025        | 0.2     | 0.41 | −0.045                 | 0.0018  |
| mRNA splicing                     | −0.026        | 0.21    | 0.41 | −0.047                 | 0.0023  |
| translation                       | 0.004         | 0.81    | 0.88 | −0.012                 | 0.32    |
| rRNA processing                   | 0.006         | 0.72    | 0.88 | −0.009                 | 0.43    |
| proteasome                        | −0.023        | 0.00085 | 0.02 | −0.025                 | 0.00017 |
| autophagy                         | −0.015        | 0.047   | 0.23 | −0.020                 | 0.0046  |
| chaperone / HSF1                  | −0.013        | 0.13    | 0.33 | −0.016                 | 0.069   |
| unfolded protein response         | −0.007        | 0.54    | 0.8  | −0.019                 | 0.03    |
| DNA double-strand repair          | −0.000        | 0.99    | 0.99 | −0.027                 | 0.0027  |
| base excision repair              | −0.001        | 0.95    | 0.99 | −0.027                 | 0.014   |
| telomere maintenance              | −0.011        | 0.63    | 0.87 | −0.037                 | 0.0044  |
| chromatin organisation            | −0.021        | 0.036   | 0.23 | −0.026                 | 0.0034  |
| mitochondrial translation         | −0.036        | 0.075   | 0.27 | −0.047                 | 0.011   |
| respiratory electron<br>transport | −0.005        | 0.77    | 0.88 | −0.021                 | 0.18    |
| TCA cycle                         | 0.018         | 0.31    | 0.51 | 0.000                  | 0.98    |
| interferon signalling             | −0.002        | 0.74    | 0.88 | −0.006                 | 0.17    |
| NF-κB / TNF                       | −0.034        | 0.0016  | 0.02 | −0.024                 | 0.0042  |
| extracellular matrix              | 0.014         | 0.12    | 0.33 | 0.017                  | 0.063   |
| collagen formation                | 0.017         | 0.12    | 0.33 | 0.015                  | 0.16    |
| insulin / IGF signalling          | −0.008        | 0.19    | 0.41 | −0.007                 | 0.23    |
| mTOR signalling                   | −0.011        | 0.3     | 0.51 | −0.016                 | 0.12    |
| cellular senescence               | −0.014        | 0.069   | 0.27 | −0.022                 | 0.0004  |
| interleukin signalling            | −0.010        | 0.025   | 0.21 | −0.013                 | 0.0013  |
| cell cycle (positive control)     | 0.006         | 0.75    | 0.88 | −0.021                 | 0.0018  |

*Skin, sun-exposed* (739 donors).

| programme           | per<br>decade | P      | FDR   | per<br>decade<br>(adj) | P (adj) |
|---------------------|---------------|--------|-------|------------------------|---------|
| splicing 96         | −0.032        | 0.037  | 0.085 | −0.024                 | 0.1     |
| pre-mRNA processing | −0.014        | 0.2    | 0.24  | −0.010                 | 0.33    |
| mRNA splicing       | −0.012        | 0.31   | 0.35  | −0.009                 | 0.46    |
| translation         | −0.046        | 0.0024 | 0.029 | −0.038                 | 0.0086  |

| programme                         | per<br>decade | P      | FDR   | per<br>decade<br>(adj) | P (adj) |
|-----------------------------------|---------------|--------|-------|------------------------|---------|
| rRNA processing                   | −0.037        | 0.03   | 0.075 | −0.029                 | 0.079   |
| proteasome                        | −0.012        | 0.072  | 0.12  | −0.009                 | 0.14    |
| autophagy                         | −0.025        | 0.0042 | 0.031 | −0.023                 | 0.0078  |
| chaperone / HSF1                  | −0.029        | 0.013  | 0.051 | −0.025                 | 0.029   |
| unfolded protein response         | −0.012        | 0.12   | 0.15  | −0.012                 | 0.13    |
| DNA double-strand repair          | −0.016        | 0.077  | 0.12  | −0.005                 | 0.49    |
| base excision repair              | −0.006        | 0.57   | 0.57  | 0.006                  | 0.44    |
| telomere maintenance              | −0.018        | 0.068  | 0.12  | −0.006                 | 0.42    |
| chromatin organisation            | −0.016        | 0.016  | 0.051 | −0.014                 | 0.034   |
| mitochondrial translation         | −0.043        | 0.0015 | 0.029 | −0.037                 | 0.0056  |
| respiratory electron<br>transport | −0.026        | 0.063  | 0.12  | −0.020                 | 0.15    |
| TCA cycle                         | −0.025        | 0.078  | 0.12  | −0.020                 | 0.16    |
| interferon signalling             | −0.007        | 0.33   | 0.35  | −0.003                 | 0.67    |
| NF-κB / TNF                       | −0.016        | 0.12   | 0.15  | −0.014                 | 0.17    |
| extracellular matrix              | 0.029         | 0.0067 | 0.033 | 0.027                  | 0.014   |
| collagen formation                | 0.027         | 0.018  | 0.051 | 0.027                  | 0.019   |
| insulin / IGF signalling          | −0.011        | 0.089  | 0.13  | −0.012                 | 0.059   |
| mTOR signalling                   | −0.029        | 0.018  | 0.051 | −0.025                 | 0.04    |
| cellular senescence               | −0.006        | 0.2    | 0.24  | −0.004                 | 0.38    |
| interleukin signalling            | 0.006         | 0.34   | 0.35  | 0.007                  | 0.27    |
| cell cycle (positive control)     | −0.023        | 0.005  | 0.031 | −0.012                 | 0.044   |

*Skin, not exposed* (626 donors).

| programme           | per<br>decade | P    | FDR  | per<br>decade<br>(adj) | P<br>(adj) |
|---------------------|---------------|------|------|------------------------|------------|
| splicing 96         | −0.023        | 0.16 | 0.78 | −0.024                 | 0.14       |
| pre-mRNA processing | −0.001        | 0.93 | 0.97 | −0.001                 | 0.91       |
| mRNA splicing       | −0.002        | 0.84 | 0.95 | −0.003                 | 0.83       |
| translation         | −0.014        | 0.37 | 0.78 | −0.015                 | 0.32       |
| rRNA processing     | −0.025        | 0.18 | 0.78 | −0.026                 | 0.16       |
| proteasome          | 0.002         | 0.71 | 0.91 | 0.002                  | 0.73       |
| autophagy           | −0.013        | 0.16 | 0.78 | −0.013                 | 0.16       |

| programme                      | per<br>decade | P     | FDR  | per<br>decade<br>(adj) | P<br>(adj) |
|--------------------------------|---------------|-------|------|------------------------|------------|
| chaperone / HSF1               | −0.023        | 0.09  | 0.78 | −0.023                 | 0.083      |
| unfolded protein response      | −0.018        | 0.053 | 0.78 | −0.017                 | 0.057      |
| DNA double-strand repair       | 0.008         | 0.37  | 0.78 | 0.005                  | 0.45       |
| base excision repair           | 0.008         | 0.46  | 0.78 | 0.005                  | 0.58       |
| telomere maintenance           | −0.003        | 0.76  | 0.91 | −0.006                 | 0.48       |
| chromatin organisation         | −0.007        | 0.35  | 0.78 | −0.007                 | 0.33       |
| mitochondrial translation      | −0.013        | 0.37  | 0.78 | −0.014                 | 0.31       |
| respiratory electron transport | 0.006         | 0.69  | 0.91 | 0.004                  | 0.77       |
| TCA cycle                      | 0.013         | 0.4   | 0.78 | 0.011                  | 0.46       |
| interferon signalling          | 0.001         | 0.95  | 0.97 | 0.000                  | 0.99       |
| NF-κB / TNF                    | −0.005        | 0.65  | 0.91 | −0.005                 | 0.68       |
| extracellular matrix           | −0.006        | 0.63  | 0.91 | −0.006                 | 0.65       |
| collagen formation             | −0.011        | 0.38  | 0.78 | −0.011                 | 0.38       |
| insulin / IGF signalling       | −0.008        | 0.26  | 0.78 | −0.008                 | 0.28       |
| mTOR signalling                | −0.015        | 0.26  | 0.78 | −0.015                 | 0.24       |
| cellular senescence            | −0.004        | 0.47  | 0.78 | −0.004                 | 0.43       |
| interleukin signalling         | −0.000        | 0.97  | 0.97 | −0.000                 | 0.99       |
| cell cycle (positive control)  | 0.003         | 0.77  | 0.91 | −0.000                 | 0.96       |

*Skeletal muscle* (839 donors).

| programme                 | per<br>decade | P       | FDR    | per<br>decade<br>(adj) | P (adj) |
|---------------------------|---------------|---------|--------|------------------------|---------|
| splicing 96               | 0.012         | 0.31    | 0.54   | 0.016                  | 0.19    |
| pre-mRNA processing       | 0.008         | 0.41    | 0.64   | 0.011                  | 0.22    |
| mRNA splicing             | 0.007         | 0.48    | 0.7    | 0.011                  | 0.25    |
| translation               | −0.037        | 0.0018  | 0.0075 | −0.031                 | 0.0097  |
| rRNA processing           | −0.016        | 0.25    | 0.49   | −0.011                 | 0.42    |
| proteasome                | −0.023        | 0.00024 | 0.0022 | −0.017                 | 0.0051  |
| autophagy                 | −0.022        | 0.0033  | 0.0092 | −0.015                 | 0.032   |
| chaperone / HSF1          | −0.011        | 0.32    | 0.54   | 0.002                  | 0.88    |
| unfolded protein response | −0.010        | 0.15    | 0.31   | −0.008                 | 0.26    |
| DNA double-strand repair  | −0.000        | 0.98    | 0.99   | −0.001                 | 0.86    |

| programme                      | per decade | P       | FDR    | per decade (adj) | P (adj) |
|--------------------------------|------------|---------|--------|------------------|---------|
| base excision repair           | 0.020      | 0.012   | 0.029  | 0.012            | 0.11    |
| telomere maintenance           | 0.001      | 0.86    | 0.99   | -0.002           | 0.76    |
| chromatin organisation         | -0.000     | 0.99    | 0.99   | 0.005            | 0.37    |
| mitochondrial translation      | -0.053     | 0.00062 | 0.0039 | -0.046           | 0.003   |
| respiratory electron transport | -0.049     | 0.0029  | 0.0091 | -0.042           | 0.011   |
| TCA cycle                      | -0.036     | 0.034   | 0.077  | -0.029           | 0.084   |
| interferon signalling          | 0.000      | 0.96    | 0.99   | 0.000            | 0.94    |
| NF- $\kappa$ B / TNF           | -0.024     | 0.00026 | 0.0022 | -0.017           | 0.0065  |
| extracellular matrix           | 0.029      | 0.0019  | 0.0075 | 0.024            | 0.011   |
| collagen formation             | 0.033      | 0.0021  | 0.0075 | 0.026            | 0.015   |
| insulin / IGF signalling       | -0.000     | 0.99    | 0.99   | -0.000           | 0.98    |
| mTOR signalling                | -0.037     | 0.00027 | 0.0022 | -0.029           | 0.0033  |
| cellular senescence            | -0.000     | 0.93    | 0.99   | 0.002            | 0.66    |
| interleukin signalling         | -0.002     | 0.75    | 0.99   | -0.001           | 0.91    |
| cell cycle (positive control)  | -0.003     | 0.59    | 0.83   | -0.005           | 0.31    |

**(c) Range restriction (post hoc).** Spread of the proliferation score, the partial correlation of the splicing score with it, and the regression slope of the splicing score on it with its 95% confidence interval, which a narrower spread does not affect; for the preregistered 96-gene set and for the 177-gene set.

| tissue (donors)            | genes in set | SD of proliferation | partial rho | P       | slope | 95% CI of slope |
|----------------------------|--------------|---------------------|-------------|---------|-------|-----------------|
| cultured fibroblasts (492) | 96           | 0.956               | 0.739       | 1.2e-85 | 0.554 | 0.509 to 0.599  |
| cultured fibroblasts (492) | 177          | 0.956               | 0.766       | 1.3e-95 | 0.569 | 0.526 to 0.611  |
| skin, sun-exposed (739)    | 96           | 0.752               | 0.225       | 6.6e-10 | 0.167 | 0.120 to 0.215  |
| skin, sun-exposed (739)    | 177          | 0.752               | 0.203       | 2.9e-08 | 0.118 | 0.079 to 0.156  |
| skin, not exposed (626)    | 96           | 0.743               | 0.056       | 0.17    | 0.072 | 0.020 to 0.125  |

| tissue (donors)         | genes in set | SD of proliferation | partial rho | P       | slope  | 95% CI of slope  |
|-------------------------|--------------|---------------------|-------------|---------|--------|------------------|
| skin, not exposed (626) | 177          | 0.743               | 0.021       | 0.6     | 0.037  | −0.005 to 0.079  |
| skeletal muscle (839)   | 96           | 0.397               | −0.115      | 0.00085 | −0.127 | −0.203 to −0.051 |
| skeletal muscle (839)   | 177          | 0.397               | −0.130      | 0.00016 | −0.139 | −0.206 to −0.072 |

The partial correlation expected in each tissue if the narrower spread of proliferation were the only difference from culture, against the one observed (96-gene set).

| tissue            | SD relative to culture | rho expected from range restriction alone | rho observed |
|-------------------|------------------------|-------------------------------------------|--------------|
| skin, sun-exposed | 0.786                  | 0.653                                     | 0.225        |
| skin, not exposed | 0.776                  | 0.649                                     | 0.056        |
| skeletal muscle   | 0.415                  | 0.415                                     | −0.115       |

Cultured fibroblasts subsampled to the spread of proliferation in sun-exposed skin: a window centred on the median, and kernel weights that down-weight the tails without discarding them; the median partial correlation over the draws with its 2.5th and 97.5th percentiles.

| scheme | target SD | cultures available | draws | SD achieved (median) | partial rho (median) | 2.5th to 97.5th percentile |
|--------|-----------|--------------------|-------|----------------------|----------------------|----------------------------|
| window | 0.752     | 338                | 500   | 0.751                | 0.872                | 0.834 to 0.910             |
| kernel | 0.752     | 492                | 500   | 0.801                | 0.770                | 0.701 to 0.857             |

**(d) Adjustment for cell composition (post hoc).** The splicing-proliferation partial correlation with the technical covariates only, and with the marker scores named in the third column added to them. Block (f) repeats the adjustment across programmes, which is what separates a composition effect on splicing from a composition effect on the whole proliferation axis.

| tissue<br>(donors)               | genes<br>in<br>set | marker<br>scores added                                | partial<br>rho,<br>technical | P       | partial<br>rho,<br>with<br>composition | P<br>(composition) |
|----------------------------------|--------------------|-------------------------------------------------------|------------------------------|---------|----------------------------------------|--------------------|
| cultured<br>fibroblasts<br>(492) | 96                 | fibroblast,<br>immune                                 | 0.739                        | 1.2e-85 | 0.722                                  | 1.4e-79            |
| cultured<br>fibroblasts<br>(492) | 177                | fibroblast,<br>immune                                 | 0.766                        | 1.3e-95 | 0.764                                  | 3.2e-94            |
| skin, sun-<br>exposed<br>(739)   | 96                 | keratinocyte,<br>fibroblast,<br>immune                | 0.225                        | 6.6e-10 | 0.101                                  | 0.0064             |
| skin, sun-<br>exposed<br>(739)   | 177                | keratinocyte,<br>fibroblast,<br>immune                | 0.203                        | 2.9e-08 | 0.123                                  | 0.00088            |
| skin, not<br>exposed<br>(626)    | 96                 | keratinocyte,<br>fibroblast,<br>immune                | 0.056                        | 0.17    | -0.058                                 | 0.15               |
| skin, not<br>exposed<br>(626)    | 177                | keratinocyte,<br>fibroblast,<br>immune                | 0.021                        | 0.6     | -0.047                                 | 0.25               |
| skeletal<br>muscle (839)         | 96                 | myofibre,<br>satellite cell,<br>fibroblast,<br>immune | -0.115                       | 0.00085 | -0.040                                 | 0.25               |
| skeletal<br>muscle (839)         | 177                | myofibre,<br>satellite cell,<br>fibroblast,<br>immune | -0.130                       | 0.00016 | -0.056                                 | 0.11               |

**(e) Positive control for donor concordance (post hoc).** Spearman correlation between a donor's cultured fibroblasts and the same donor's sun-exposed skin for genes whose expression depends strongly on genotype or on sex. The column before the note is the product of the gene's correlations with sex in the two sample types, that is, the concordance that sex alone would produce. For comparison, the preregistered exploratory analysis gave  $\rho = -0.10$  to  $+0.13$  for the 25 programme scores in the same 388 donors (Supplementary Fig. S8c).

| gene        | group                  | don<br>ors | rho   | P       | rho<br>expec<br>ted<br>from<br>sex<br>alone | note                                                                 |
|-------------|------------------------|------------|-------|---------|---------------------------------------------|----------------------------------------------------------------------|
| XIST        | sex<br>chromosome      | 388        | 0.730 | 1e-65   | 0.664                                       |                                                                      |
| RPS4Y1      | sex<br>chromosome      | 388        | 0.666 | 5.5e-51 | 0.670                                       |                                                                      |
| DDX3Y       | sex<br>chromosome      | 388        | 0.729 | 1.8e-65 | 0.671                                       |                                                                      |
| KDM5D       | sex<br>chromosome      | 388        | 0.692 | 1.5e-56 | 0.670                                       |                                                                      |
| UTY         | sex<br>chromosome      | 388        | 0.668 | 1.6e-51 | 0.666                                       |                                                                      |
| USP9Y       | sex<br>chromosome      | 388        | 0.698 | 5.8e-58 | 0.667                                       |                                                                      |
| EIF1AY      | sex<br>chromosome      | 388        | 0.736 | 1.7e-67 | 0.668                                       |                                                                      |
| GSTM1       | genotype-<br>dependent | 388        | 0.750 | 2.9e-71 | 0.004                                       |                                                                      |
| GSTT1       | genotype-<br>dependent |            |       |         |                                             | absent from<br>the gene-<br>symbol map                               |
| UGT2B1<br>7 | genotype-<br>dependent |            |       |         |                                             | present in the<br>annotation but<br>filtered out as<br>not expressed |
| ERAP2       | genotype-<br>dependent | 388        | 0.749 | 5.1e-71 | 0.002                                       |                                                                      |

**(f) The same composition adjustment for six sets (post hoc).** Run on the splicing sets alone, block (d) cannot distinguish an adjustment that removes splicing's coupling from one that flattens the proliferation signal. Repeated for the mitotic cell-cycle positive control, collagen formation and the two unselected Reactome splicing definitions, the two separate: the cell-cycle programme and collagen formation do not move, while every definition of the splicing set falls (the values of Fig. 7d). The fibroblast marker score contains COL1A1 and COL1A2, so collagen formation is not an independent check.

| tissue                  | programme                           | partial<br>rho,<br>technical | partial rho,<br>with<br>composition | difference |
|-------------------------|-------------------------------------|------------------------------|-------------------------------------|------------|
| cultured<br>fibroblasts | 96                                  | 0.739                        | 0.722                               | 0.017      |
| cultured<br>fibroblasts | 177                                 | 0.766                        | 0.764                               | 0.003      |
| cultured<br>fibroblasts | pre-mRNA<br>processing              | 0.691                        | 0.653                               | 0.038      |
| cultured<br>fibroblasts | mRNA splicing                       | 0.689                        | 0.651                               | 0.038      |
| cultured<br>fibroblasts | collagen<br>formation               | 0.049                        | 0.422                               | -0.373     |
| cultured<br>fibroblasts | cell cycle<br>(positive<br>control) | 0.941                        | 0.952                               | -0.011     |
| skin, sun-<br>exposed   | 96                                  | 0.225                        | 0.101                               | 0.124      |
| skin, sun-<br>exposed   | 177                                 | 0.203                        | 0.123                               | 0.080      |
| skin, sun-<br>exposed   | pre-mRNA<br>processing              | 0.145                        | 0.082                               | 0.062      |
| skin, sun-<br>exposed   | mRNA splicing                       | 0.134                        | 0.063                               | 0.071      |
| skin, sun-<br>exposed   | collagen<br>formation               | 0.030                        | 0.027                               | 0.003      |
| skin, sun-<br>exposed   | cell cycle<br>(positive<br>control) | 0.670                        | 0.666                               | 0.004      |
| skin, not<br>exposed    | 96                                  | 0.056                        | -0.058                              | 0.114      |
| skin, not<br>exposed    | 177                                 | 0.021                        | -0.047                              | 0.068      |
| skin, not<br>exposed    | pre-mRNA<br>processing              | -0.025                       | -0.074                              | 0.049      |
| skin, not<br>exposed    | mRNA splicing                       | -0.044                       | -0.094                              | 0.049      |
| skin, not<br>exposed    | collagen<br>formation               | 0.086                        | 0.110                               | -0.024     |
| skin, not<br>exposed    | cell cycle<br>(positive             | 0.597                        | 0.586                               | 0.011      |

| tissue             | programme                           | partial<br>rho,<br>technical | partial rho,<br>with<br>composition | difference |
|--------------------|-------------------------------------|------------------------------|-------------------------------------|------------|
|                    | control)                            |                              |                                     |            |
| skeletal<br>muscle | 96                                  | -0.115                       | -0.040                              | -0.075     |
| skeletal<br>muscle | 177                                 | -0.130                       | -0.056                              | -0.074     |
| skeletal<br>muscle | pre-mRNA<br>processing              | -0.131                       | -0.026                              | -0.105     |
| skeletal<br>muscle | mRNA splicing                       | -0.139                       | -0.037                              | -0.102     |
| skeletal<br>muscle | collagen<br>formation               | 0.240                        | 0.031                               | 0.210      |
| skeletal<br>muscle | cell cycle<br>(positive<br>control) | 0.031                        | 0.157                               | -0.126     |

**(g) What a two-compartment tissue should show (post hoc).** Between-donor variation in each score written as a part that follows the fraction of cycling cells and a part that does not. The proliferation score and the mitotic cell-cycle programme are both read from cycling cells, so their correlation fixes how well the design sees that fraction; if a share  $w$  of the splicing score's variation followed it, the expected correlation would be  $w$  times that value. The row without a  $w$  is the observed pair.

| share $w$ | expected splicing<br>rho | expected cell-cycle<br>rho |
|-----------|--------------------------|----------------------------|
| 0.10      | 0.067                    | 0.667                      |
| 0.20      | 0.131                    | 0.665                      |
| 0.30      | 0.194                    | 0.667                      |
| 0.40      | 0.261                    | 0.663                      |
| 0.50      | 0.329                    | 0.666                      |
| 0.60      | 0.394                    | 0.666                      |
| 0.70      | 0.460                    | 0.668                      |
| 0.80      | 0.528                    | 0.667                      |
| 0.90      | 0.597                    | 0.665                      |
| 1.00      | 0.666                    | 0.668                      |
|           | 0.225                    | 0.670                      |

# Supplementary Note 1. Preregistration for the GTEx analysis

## What this document is

The GTEx analysis reported in Fig. 7 was planned before the data were seen. The plan below was written in Korean on 13 September 2026, and its SHA-256 digest was recorded in `public_data_tierA/logs/preregistration_sha256.txt` before any GTEx expression file was downloaded or opened; the analysis script `scripts/run_gtex_boundary.R` recomputes the digest and stops if it does not match. The digest is

61972b991dc6de770478cd7f6df39a5319e11107ef2993aac2faeee8eb8b774e

and it still matches the current file, which has not been amended since it was locked.

This was an internal record. It was **not** deposited with a public registry such as OSF or AsPredicted, so a reader cannot verify independently when it was written; what can be verified is that the document the analysis ran against is the document reproduced here, and that its stated criteria are the criteria against which the outcomes below are reported, including the two that the analysis failed. The translation is complete rather than summarised. Gene names, dataset identifiers, statistical terms and numbers are unchanged; where the Korean is ambiguous it is translated literally and a bracketed translator's note is added.

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## Translation

### Preregistration: culture-versus-tissue boundary test (GTEx v8 via recount3)

- Written: 2026-09-13
- Status: **written and locked before the GTEx expression data were downloaded or opened**
- Locking method: record the SHA-256 of this file in `public_data_tierA/logs/preregistration_sha256.txt`, then run the analysis script.

## 1. Purpose

The manuscript's present claim is as follows. In cultured human fibroblasts, splicing-factor expression and eight ageing programmes (chromatin organisation, cellular senescence, pre-mRNA processing, mRNA splicing, interferon signalling, DNA double-strand break repair, telomere

maintenance, base excision repair) are readouts of division rate, whereas collagen formation and the unfolded protein response are not. The limitations section stated that “it says nothing about post-mitotic tissue”. This test establishes that boundary from data rather than by assumption. GTEx is the only public resource in which **cultured fibroblasts**, **skin tissue** and **skeletal muscle (post-mitotic tissue)** can be compared under one definition in the same donor population.

## 2. Data (processed only)

| File                                                    | Content                                                                                                          | Size         |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------|
| gtex.gene_sums.SKIN<br>.G026.gz + 3<br>metadata files   | Cells - Cultured<br>fibroblasts, Skin -<br>Sun Exposed (Lower<br>leg), Skin - Not Sun<br>Exposed<br>(Suprapubic) | about 196 MB |
| gtex.gene_sums.MUSC<br>LE.G026.gz + 3<br>metadata files | Muscle - Skeletal                                                                                                | about 81 MB  |

About 277 MB in total. FASTQ and junction files will not be downloaded. Splicing **outcome** (unannotated junctions) will therefore not be computed in GTEx, and only programme-level expression analysis will be performed.

## 3. Definitions fixed before seeing results

- **Gene ID → symbol:** use the mapping already used in the project (Geneid → Gene\_name from GSE282054\_raw\_counts.txt.gz), unchanged.
- **Normalisation:** separately per tissue type. edgeR filterByExpr → TMM → log<sub>2</sub> CPM → gene-wise z.
- **Programme scores:** the list of 24 Reactome sets in scripts/run\_hallmark\_audit.R, unchanged. Mean z of the genes expressed in that tissue; excluded if fewer than 15.
- **Splicing set:** the 96 genes in conserved\_core/age\_down\_splicing\_core.txt.
- **Proliferation score:** mean z of the 20 markers in scripts/run\_outcome\_vs\_expression.R (MKI67, CCNB1, CCNA2, CDK1, TOP2A, BIRC5, BUB1, PLK1, AURKA, TYMS, RRM2, PCNA, MCM2-MCM7, TK1, UBE2C).
- **Prevention of circularity:** when a programme is compared with the proliferation score or adjusted for proliferation, the 20 proliferation markers contained in that programme are first removed from the programme score. For the mitotic cell cycle set (the positive control), the adjusted result is not interpreted.

- **Age:** the midpoint of GTEx's released age bands (20–29, ..., 70–79), i.e. 25, 35, ..., 75.
- **Technical covariates:** RIN, ischaemic time,  $\log_{10}$  library size. Fields not present in the recount3 metadata will not be used, and that fact will be recorded.
- **Biological covariates:** sex, Hardy scale of death circumstances (missing values as a separate category).
- **Sample exclusions:** samples not belonging to the four tissue types above; missing age; the lowest 1% of library size within a tissue type. Where one donor has several samples of one tissue type, the one with the largest library size.
- **Definition of the correlation:** the primary measure is the **partial Spearman  $\rho$**  after regressing the technical covariates out of both variables (GTEx is post-mortem and surgical material, so technical variation is large). The raw Spearman  $\rho$  is a sensitivity analysis.
- **Age model:**  $\text{score} \sim \text{age} + \text{sex} + \text{Hardy} + \text{RIN} + \text{ischaemic time} + \log_{10} \text{ library size}$ ; the proliferation-adjusted model adds the proliferation score to this.  $\text{Attenuation} = 1 - \beta_{\text{adj}}/\beta$ .

#### 4. Primary hypotheses and decision criteria

|                                                    | Hypothesis                                                                                            | Success                                     | Failure                                                                |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------|
| <b>H1</b> culture replication                      | Partial $\rho$ between the splicing-96 score and the proliferation score in GTEx cultured fibroblasts | $\rho \geq 0.5$                             | $\rho < 0.3$ (0.3–0.5 is partial replication)                          |
| <b>H2</b> coupling weaker in tissue                | The splicing-96–proliferation $\rho$ in lower-leg skin tissue is smaller than in cultured fibroblasts | Fisher $z$ difference, one-sided $P < 0.05$ | no difference, or the opposite                                         |
| <b>H3</b> post-mitotic tissue boundary             | Partial $\rho$ between splicing-96 and proliferation in skeletal muscle                               | $\rho < 0.3$                                | $\rho \geq 0.5$                                                        |
| <b>H4</b> attenuation of the age effect in culture | In cultured fibroblasts, the age effect on splicing-96 is                                             | attenuation $\geq 50\%$                     | attenuation $< 25\%$ . If the unadjusted age $P \geq 0.05$ , report as |

|                                              | Hypothesis                                                                          | Success               | Failure                                                                                                      |
|----------------------------------------------|-------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------|
|                                              | attenuated after adjustment for proliferation                                       |                       | <b>not evaluable</b>                                                                                         |
| <b>H5</b> collagen dissociation in culture   | Partial $\rho$ between collagen formation and proliferation in cultured fibroblasts | $ \rho  < 0.3$        | $ \rho  \geq 0.5$                                                                                            |
| <b>H6</b> positive control for tissue ageing | Collagen formation falls with age in lower-leg skin tissue                          | $\beta < 0, P < 0.05$ | $\beta \geq 0$ or $P \geq 0.05$ (in which case the whole tissue age analysis is interpreted as underpowered) |

Note on H2: because the epidermis of skin contains proliferating basal keratinocytes, the coupling may persist in tissue and H2 may fail. In that case the conclusion is restricted to “skin tissue is not a post-mitotic tissue”, and the judgement about the boundary is left to H3.

## 5. Exploratory analyses (no success criteria)

- **S1** Compare the partial  $\rho$  with proliferation of the 24 programmes side by side in culture, the two skin sites and skeletal muscle (the tissue version of Fig. 8A). [Translator’s note: “Fig. 8A” is the panel number in the draft current at the time of writing; it is Fig. 6a in the revised manuscript.]
- **S2** Compare the age slope of collagen formation between sun-exposed (lower leg) and non-exposed (suprapubic) skin.
- **S3** Donor-level concordance of programme scores between a donor’s cultured fibroblasts and skin tissue.
- **S4** Skin tissue composition sensitivity: add keratinocyte marker (KRT5, KRT14, KRT1, KRT10) and dermal fibroblast marker (COL1A1, COL1A2, DCN, LUM, PDGFRA) scores as covariates.
- **S5** Full table of age effects for the 24 programmes  $\times$  tissue types, before and after adjustment for proliferation.

## 6. Multiple testing

- H1–H6: success is decided by the effect-size criteria above. The  $P$  value of each hypothesis is reported together after Benjamini–Hochberg correction within the six primary hypotheses.
- S1, S5: within each tissue type, the proliferation family and the age family are BH-corrected separately (as in the existing audit).

## 7. Reporting principles

- Report both successes and failures. Failed hypotheses will not be removed from the manuscript.
- Do not change definitions, thresholds or sample-exclusion rules after seeing results. If a change is unavoidable, record the date and reason in the amendment history below and report **the result under the original definition as well**.
- The material analysed is reprocessed public data and no causal claim is made. GTEx age is given in ten-year bands, so age effects are interpreted only at band resolution.

## Amendment history

(none)

## Outcomes

Recorded in results\_gtex\_culture\_tissue\_boundary\_ko.md. Final samples after the preregistered exclusions: cultured fibroblasts 511 donors (492 with complete technical covariates), lower-leg skin 739, suprapubic skin 626, skeletal muscle 839; 388 donors contributed both cultured fibroblasts and lower-leg skin.

|           | Preregistered criterion                            | Estimate obtained                                                            | <i>P</i>                          | BH-FDR                | Verdict          |
|-----------|----------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------|-----------------------|------------------|
| <b>H1</b> | $\rho \geq 0.5$ succeeds;<br>$\rho < 0.3$ fails    | partial $\rho = +0.739$ in cultured fibroblasts ( $n = 492$ )                | $1.2 \times 10^{-85}$             | $7.1 \times 10^{-85}$ | <b>supported</b> |
| <b>H2</b> | one-sided $P < 0.05$ for the Fisher $z$ difference | lower-leg skin $+0.225$ ( $n = 739$ ) versus culture $+0.739$ ; $z = -11.95$ | $3.2 \times 10^{-33}$ (one-sided) | $9.6 \times 10^{-33}$ | <b>supported</b> |
| <b>H3</b> | $\rho < 0.3$ succeeds;<br>$\rho \geq 0.5$          | partial $\rho = -0.115$ in                                                   | $8.5 \times 10^{-4}$              | $1.7 \times 10^{-3}$  | <b>supported</b> |

|           | Preregistered criterion                                                                    | Estimate obtained                                                     | <i>P</i> | BH-FDR | Verdict                                   |
|-----------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------|--------|-------------------------------------------|
|           | fails                                                                                      | skeletal muscle ( <i>n</i> = 839)                                     |          |        |                                           |
| <b>H4</b> | attenuation $\geq 50\%$ succeeds; not evaluable if the unadjusted age <i>P</i> $\geq 0.05$ | unadjusted age effect $-0.025$ per decade                             | 0.33     | 0.33   | <b>not evaluable</b>                      |
| <b>H5</b> | $ \rho  < 0.3$ succeeds; $ \rho  \geq 0.5$ fails                                           | partial $\rho = +0.049$ for collagen formation in culture             | 0.28     | 0.33   | <b>supported</b>                          |
| <b>H6</b> | $\beta < 0$ and <i>P</i> $< 0.05$ succeeds                                                 | collagen formation in lower-leg skin $+0.027$ per decade, i.e. rising | 0.018    | 0.027  | <b>not supported (opposite direction)</b> |

Because H6 failed, the preregistered rule was applied: age effects in the GTEx tissues are treated as not validated by a positive control and are not used to support any claim in the manuscript. This applies to the tissue columns of S5 and to the S2 comparison of sun-exposed with non-exposed skin. It was noted at the time, without changing that rule, that the failure was a significant effect in the opposite direction rather than an absence of power.

## Deviations and post hoc additions

In the order in which they occurred. The preregistration's own amendment history is empty; the first two items below are recorded in the outcome record, and the rest are later additions, each labelled post hoc.

1. **Before the analysis was run — the  $P$  value attached to H4.** The preregistration did not state which  $P$  value belonged to H4. The draft script attached the proliferation-adjusted age  $P$ ; before any result was seen this was changed to the **unadjusted** age  $P$ , which is the quantity the preregistered rule uses to decide whether H4 is evaluable. The decision rule itself was unchanged. Recorded in `results_gtex_culture_tissue_boundary_ko.md`.
2. **Before the analysis was run — a failed first execution.** The first run stopped before R started because the output directory did not exist. The directory was created and the same script was run again. No definition changed.
3. **The splicing set is the 96-gene version, fixed by the preregistration.** After the preregistration was locked, a fourth donor cohort (GSE226189) was removed from the study on 2026-09-13 because its processing batches coincide with age strata, which changed the study-wide splicing set from 96 genes to 177. The preregistered verdicts above stand on the 96-gene file, which was not rewritten. Repeating the same analysis with the 177-gene set is a **post hoc sensitivity analysis**: cultured fibroblasts +0.766, lower-leg skin +0.203, skeletal muscle −0.130, and the culture age effect −0.024 per decade unadjusted ( $P = 0.35$ ) and −0.052 adjusted ( $P = 0.001$ ). Every verdict is the same and H4 remains not evaluable.
4. **Post hoc — 32 off-trend cultures.** After the outcomes were known, 32 of the 511 cultures were found to lie off the main trend at low proliferation; they are enriched in particular RNA-extraction and sequencing batches and have lower RIN. Excluding them raises the raw correlation from 0.72 to 0.89. The preregistered estimate includes them, and the exclusion is reported only as an observation.
5. **Post hoc — range restriction and variance-matched subsampling** (2026-09-15, `results_gtex_posthoc_ko.md`, `public_data_tierA/derived/gtex_posthoc/`). Added in response to the objection that proliferation varies less between tissue donors than between cultures. The regression slope, which range restriction does not affect, falls from 0.554 in culture to 0.167 in lower-leg skin, 0.072 in suprapubic skin and −0.127 in muscle; a Thorndike case-II correction predicts  $\rho = 0.653$  for lower-leg skin against the 0.225 observed; and cultures subsampled to the skin's proliferation spread

give a median partial  $\rho$  of 0.872 (95% range 0.834–0.910) with a median-centred window and 0.770 (0.701–0.857) with the tails down-weighted rather than discarded.

6. **Post hoc — adjustment for cell composition** (2026-09-15). This extends the preregistered exploratory analysis S4 in three ways, all of them after the outcomes were known: the marker panels were enlarged (keratinocyte KRT5, KRT14, KRT1, KRT10, TP63, IVL, LOR; fibroblast COL1A1, COL1A2, DCN, LUM, PDGFRA, **PDGFRB**; immune PTPRC; and for muscle myofibre ACTA1, MYH1, MYH2, MYH7, CKM and satellite-cell PAX7, MYF5); the fibroblast panel thus keeps the five genes the preregistration named, PDGFRA among them, and **adds PDGFRB**; and the adjustment was applied to the splicing-proliferation coupling itself, whereas S4 as written applied it to age effects. With composition in the covariates the partial  $\rho$  between the splicing-96 score and the proliferation score falls from +0.225 ( $P = 6.6 \times 10^{-10}$ ) to +0.101 ( $P = 0.006$ ) in lower-leg skin ( $n = 739$  donors), from +0.056 ( $P = 0.17$ ) to  $-0.058$  ( $P = 0.15$ ) in suprapubic skin ( $n = 626$ ), and from  $-0.115$  ( $P = 8.5 \times 10^{-4}$ ) to  $-0.040$  ( $P = 0.25$ ) in muscle ( $n = 839$ ), while culture is essentially unchanged (+0.739 to +0.722,  $n = 492$ ; `public_data_tierA/derived/gtex_posthoc/composition_adjusted.tsv`). The fibroblast panel was revised once, on the same day and after the first post hoc run: that run had PDGFRB in place of the preregistered PDGFRA and gave +0.111 ( $P = 0.003$ ) in lower-leg skin,  $-0.052$  ( $P = 0.19$ ) in suprapubic skin,  $-0.021$  ( $P = 0.55$ ) in muscle and +0.721 in culture, after which PDGFRA was restored and PDGFRB kept as an addition, with no other change to markers, covariates, samples or model; the values reported above and in the main text are from the revised panel, and the first-run table is kept as `composition_adjusted_first_definition.tsv`. The preregistered S4 result on the age effect, which was run with the preregistered panel, stands as recorded (lower-leg skin splicing-96 age effect  $-0.032$ ,  $P = 0.037$ , becoming  $-0.013$ ,  $P = 0.32$ , with keratinocyte and fibroblast markers), and is not used as evidence, since it is a tissue age effect.
7. **Post hoc — genotype-driven positive control for donor concordance** (2026-09-15). The preregistered exploratory analysis S3 found no donor-level concordance between cultured fibroblasts and skin for any of the 25 programme scores ( $\rho = -0.10$  to +0.13, none at  $FDR < 0.05$ ). To establish that the design can detect donor identity at all, genes whose expression is strongly genotype-dependent were tested in the same 388 donors and the same matrices: *GSTM1*  $\rho = +0.750$  ( $P = 3 \times 10^{-71}$ ) and *ERAP2*  $\rho = +0.749$  ( $P = 5 \times 10^{-71}$ ), neither explained by sex, together with seven sex-chromosome genes at  $\rho = 0.666$  to 0.736. This control was not preregistered.

8. **Post hoc — redefinition of the comparison cohort** (2026-09-15). Fig. 7c compares the GTEx cultures with the fibroblast donor cohort GSE113957. That cohort was redefined after the outcomes were known, on grounds independent of the GTEx analysis: ten progeria donors (nine of them in the 142-sample analysed set) and 26 donors under 20 years of age were removed, and cell repository, sequencing instrument and sex were added as covariates, leaving 107 normal adults. For the preregistered 96-gene set this changes the comparison cohort's estimates from  $-0.093$  per decade unadjusted and  $-0.031$  adjusted, as recorded at the time, to  $-0.128$  unadjusted ( $P = 4 \times 10^{-6}$ ) and  $-0.033$  adjusted (95% CI  $-0.064$  to  $-0.002$ ,  $P = 0.04$ ). Nothing in the GTEx analysis itself was recomputed.
9. **Post hoc — combined residual age effect** (2026-09-16, `public_data_tierA/derived/residual_meta/residual_age_effect_meta.tsv`). The proliferation-adjusted age effects on the preregistered 96-gene set in the GTEx cultures ( $-0.053$  per decade,  $n = 492$  donors) and in the 107-donor cohort ( $-0.033$ ) were combined by inverse variance:  $-0.042$  per decade (95% CI  $-0.065$  to  $-0.019$ , 599 donors). Unadjusted estimates were not combined. This was not preregistered; it uses the cultures only, not the tissues to which the rule that followed the failure of H6 applies. Withdrawn at revision (2026-09-18): the pooled value is no longer reported anywhere in the manuscript or its tables; the two estimates are shown side by side in Fig. 7c and Supplementary Table 5 and are not combined, because one is an attenuation and the other a suppression effect in cultures with no unadjusted age effect, and with two studies the between-study variance is not estimable.
10. **Post hoc — specificity of the coupling in GTEx cultures** (2026-09-16, `public_data_tierA/derived/machinery_specificity/gtex_culture_partials.tsv`; Supplementary Fig. S5b, Supplementary Table 5). The comparison of splicing definitions with translation, Metabolism of RNA and rRNA processing, made first against the counted division rate, was repeated against the 20-marker proliferation score in the 511 GTEx cultures. The asymmetry seen against the counted division rate was not reproduced: with the other score held constant, translation keeps a partial  $\rho$  of 0.51 to 0.59 and the splicing definitions 0.28 to 0.41. This was not preregistered, and it uses the cultures only.

Two further discrepancies are recorded for completeness. The preregistration estimated the download at about 277 MB; the files actually retrieved total 265 MB, with SHA-256 digests in `public_data_tierA/gtex/gtex_sha256.txt`. And the panel numbering

referred to in S1 is that of the draft current on 2026-09-13, not of the revised manuscript.

## Supplementary Note 2. Preregistration for the methylation clock analysis

### What this document is

The methylation analysis reported in Supplementary Fig. S7 was planned before the data were seen. The plan below was written in Korean on 16 September 2026, and its SHA-256 digest was recorded before the methylation matrix of GSE179847 was downloaded; the analysis script `scripts/revision/run_methylation_clock.R` recomputes the digest and stops if it does not match. The digest is

be51a7550c982e2071ebdb80f36ab7098965351c6e14a65434e8df275505d929

and it still matches the current file, which has not been amended since it was locked.

As with Supplementary Note 1, this was an internal record and was not deposited with a public registry, so a reader cannot verify independently when it was written; what can be verified is that the document the analysis ran against is the document reproduced here, and that its criteria are the criteria against which the outcomes below are reported, including the three the analysis failed. The translation is complete rather than summarised.

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### Translation

#### Preregistration: the relationship between methylation clocks and the transcriptomic proliferation readout

This document was written **before downloading** GSE179847 (Cellular Lifespan Study DNA methylation, EPIC array, 479 samples). The SHA-256 is recorded, then the data are downloaded, and the analysis script checks the digest before it runs. After locking, additions are made only to the amendment history at the end of the document.

#### 1. Background and the question

This study has shown that transcriptional programmes in cultured fibroblasts, splicing factors among them, are readouts of the **measured division rate**. That methylation clocks run fast in culture (Sturm 2019) and that cell division causes methylation loss in late-replicating domains

(Endicott 2022) are already known. **“The clock follows division too” is therefore not itself a new question.**

The new question is one. **In the same cultures, are the two biomarker families — the transcriptomic proliferation readout and the methylation clock — measuring the same variable?** GSE179847 and GSE179848 come from the same cell lines and the same passages, so they can be matched directly at the sample level.

## 2. Data

- **GSE179847:** GSE179847\_Cell\_lifespan\_DNA<sub>m</sub>\_processed\_matrix.csv.gz (about 2.0 GB). Rows are CpGs, columns are per-sample beta values and detection *P* values. Sample identifiers are sentrix IDs, linked to GSM and unique\_variable\_name through the GEO sample list.
- **GSE179848:** the RNA-seq data already held. 328 libraries carry a measured division rate.
- **Clock coefficients:** the Horvath 2013 multi-tissue clock (353 CpGs). The skin and blood clock (391) and PhenoAge (513) are used only if their coefficient tables can be obtained from a public source; if they cannot, that fact is recorded.

## 3. Definitions fixed in advance

- **DNA<sub>m</sub> age:** computed with each clock’s published coefficients. For Horvath 2013 the published inverse transformation (adult basis) is used as it stands. Missing CpGs are replaced by the cohort mean of that CpG and **the number replaced is reported**.
- **Clock acceleration:** the residual of DNA<sub>m</sub> age regressed on days in culture within a cell line.
- **Measured division rate:** 24 / population\_doubling\_time\_uhours\_per\_division (divisions per day). This name is not used for any other dataset.
- **Transcriptomic proliferation score:** the mean *z* of the existing 20 markers (MKI67 and the rest).
- **Splicing score:** the mean *z* of the existing 177-gene pre-mRNA processing set.
- **Matching:** only samples with exactly the same unique\_variable\_name are treated as a pair. If no exact match exists, matching is restricted to samples identical in cell line, treatment, oxygen and passage, and the number matched is reported.
- **Note on normalisation:** this analysis does not reproduce the original paper’s pipeline (BMIQ and the rest), so **absolute age estimates are approximations**. Absolute values are not interpreted; only correlations and residuals are.

#### 4. Hypotheses and success criteria

|                           | Hypothesis                                                                                                      | Success criterion                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>M1</b> (sanity check)  | In untreated healthy cell lines, DNAm age increases with cumulative divisions                                   | $\rho \geq 0.3, P < 0.05$                     |
| <b>M2</b>                 | Clock acceleration is associated with the <b>measured division rate</b>                                         | $ \rho  \geq 0.3, P < 0.05$                   |
| <b>M3</b> (the main test) | In the same cultures, the transcriptomic proliferation score is associated with DNAm age                        | $ \rho  \geq 0.3, P < 0.05$                   |
| <b>M4</b>                 | The association between the splicing score and DNAm age shrinks when the measured division rate is adjusted for | the partial correlation falls by at least 50% |

- If M1 fails, the clock computation is taken to be wrong and **M2-M4 are not interpreted**. In that case the failure is recorded as it stands.
- If any hypothesis comes out in the opposite direction it is reported as it stands. Criteria are not changed after results are seen.
- Multiple testing is corrected across the four hypotheses with Benjamini-Hochberg.

#### 5. Exclusion criteria

- Measurements with a detection  $P$  value above 0.01 are treated as missing.
- Samples missing 30% or more of the clock CpGs are excluded and their number reported.
- Samples with no division rate, or a rate of zero or below, are excluded (the same rule as the RNA analysis).

#### 6. Reporting principles

- That absolute DNAm ages are approximations is stated in the figure and in the text.
- This analysis occupies **no more than one panel and one paragraph** in the paper. The claim is limited to one thing: whether the two biomarker families read the same variable in culture.
- Endicott 2022 and Sturm 2019 are named as prior work, and what this analysis re-confirms is distinguished from what it adds.

## 7. Amendment history

(none)

## Outcomes

The analysis was run on 16 September 2026 with scripts/revision/run\_methylation\_clock.R, which verified the digest above before it started. Of 479 arrays, none was missing more than 30% of clock CpGs; the detection filter set 0.018% of measurements to missing and the 99 remaining gaps were filled with the cohort mean of the CpG. Clock CpGs available: 334 of 353 (Horvath 2013), 391 of 391 (skin and blood), 513 of 513 (PhenoAge), 63 of 69 (Hannum). After the division-rate exclusion, 471 methylation samples from nine cell lines and nineteen treatments remained, and exact matching on unique\_variable\_name gave 109 cultures with both a clock and transcriptional scores, from lines HC1, HC2 and HC4.

|    | Test                                                           | $\rho$ | $P$                | BH-FDR | n   | Criterion met |
|----|----------------------------------------------------------------|--------|--------------------|--------|-----|---------------|
| M1 | DNAm age vs cumulative doubling times, untreated healthy lines | +0.356 | $8 \times 10^{-4}$ | 0.0015 | 86  | <b>yes</b>    |
| M2 | clock acceleration vs measured division rate                   | -0.014 | 0.76               | 0.76   | 471 | no            |
| M3 | proliferation score vs DNAm age, paired cultures               | +0.162 | 0.093              | 0.124  | 109 | no            |
| M4 | splicing                                                       | -0.338 | $3 \times 10^{-4}$ | 0.0013 | 109 | no            |

| Test                                                        | $\rho$ | $P$ | BH-FDR | n | Criterion met |
|-------------------------------------------------------------|--------|-----|--------|---|---------------|
| vs<br>DNAm<br>age with<br>division<br>rate held<br>constant |        |     |        |   |               |

M1 was met, so M2–M4 are interpreted under the preregistered rule. M2 and M3 failed on the effect-size criterion fixed in advance ( $|\rho| \geq 0.3$ ), not on significance alone: the 95% confidence interval for M3 is  $-0.03$  to  $+0.34$ , so a correlation of the preregistered size is not excluded by these 109 cultures, and a larger paired set could change the verdict. M4 asked for attenuation of at least 50%; the association did not attenuate but changed sign, from  $+0.15$  unadjusted to  $-0.34$  with division rate held constant (an attenuation of  $-131\%$ ), so the criterion is not met. That sign change is recorded and not interpreted.

Supporting numbers, none of them preregistered tests. Within the 109 paired cultures the transcriptional scores follow the measured division rate as they do throughout the study (splicing  $\rho = +0.77$ , proliferation  $+0.69$ , and the two agree with each other at  $+0.82$ ), while DNAm age follows it at  $+0.41$  before time in culture is removed and at zero after. Three further clocks, computed the same way, give acceleration-rate correlations of  $+0.18$  for skin and blood ( $P = 7 \times 10^{-5}$ ),  $+0.14$  for Hannum ( $P = 0.003$ ) and  $+0.06$  for PhenoAge ( $P = 0.18$ ); all are below the preregistered  $0.3$ .

## Deviations and post hoc additions

1. **The Hannum clock was added.** The preregistration named Horvath 2013, the skin and blood clock and PhenoAge. Hannum's clock was computed as well, because its coefficient table was obtained at the same time. It is reported only as a supporting comparison in Supplementary Fig. S7d and carries none of the preregistered verdicts.
2. **How “untreated healthy cell lines” was operationalised for M1.** The preregistration did not define the phrase. It was taken as the six healthy-donor lines (HC1–HC6) under the study's own Control treatment label at 21% oxygen, which gives  $n = 86$ . The SURF1 patient lines and every treated or hypoxic culture are excluded from M1 only; M2 uses all 471.
3. **Clock acceleration was fitted with a common time slope.** “The residual of DNAm age regressed on days in culture within a cell line” was implemented as a single regression on days with a cell-line term, which gives every line the same slope. Fitting a separate slope per

line instead gives  $\rho = +0.092$  against the measured division rate ( $P = 0.045$ ), still well below the preregistered 0.3, and the correlation within individual lines ranges from  $-0.74$  to  $+0.40$ . The verdict on M2 is the same either way.

4. **Post hoc — the confidence interval reported for M3.** The preregistration fixed a threshold but did not ask for an interval. The Fisher-z interval is given in the manuscript and above so that the failure is not read as evidence of absence.
5. **Reporting limit** (recorded 2026-09-17). The preregistration allowed one panel and one paragraph. The analysis is reported as one four-panel supplementary figure (Supplementary Fig. S7: the sanity check M1, the tests M2 and M3, and the supporting clocks; M4 is not drawn, and its estimate is in Supplementary Table 8) so that every outcome is visible, with one Results paragraph, one sentence of Discussion, one Methods subsection and Supplementary Table 8. The claim remains limited as preregistered, and no claim about clocks is made.

## Supplementary Data

Machine-readable tables accompany this file.

| File                 | Deposited as                                     | Contents                                                                                                                                                                                                                                                                   |
|----------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supplementary Data 1 | SupplementaryData1_contrasts.tsv                 | The 63 contrasts: change in the 177-gene splicing set, in the cell-cycle set and in the splicing set without the 23 shared genes, with the class of each contrast (leave_out_defining_series.tsv in the same folder gives the refit without the three set-defining series) |
| Supplementary Data 2 | SupplementaryData2_gtex_programmes_by_tissue.tsv | Programme scores against proliferation and donor age in every GTEx tissue, with age effects before and after adjustment                                                                                                                                                    |
| Supplementary Data 3 | SupplementaryData3_gene_level_by_cohort.ts       | For each of the four cohort definitions: the                                                                                                                                                                                                                               |

| File                 | Deposited as                                          | Contents                                                                                                                                                              |
|----------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | v                                                     | proliferation-age correlation, the machinery-proliferation correlation, and the number of genes with an age association before and after adjustment for proliferation |
| Supplementary Data 4 | SupplementaryData4_programme_division_rate.tsv        | Division-rate correlations for every programme, with and without cell-cycle genes                                                                                     |
| Supplementary Data 5 | SupplementaryData5_residual_specification.tsv         | The residual age effect under every model specification — splicing definition, comparator programme and covariate set (Supplementary Table 5)                         |
| Supplementary Data 6 | SupplementaryData6_composition_adjusted.tsv           | Partial correlation with proliferation before and after cell-composition marker scores, for six sets in four tissues (Fig. 7d, Supplementary Fig. S8b)                |
| Supplementary Data 7 | SupplementaryData7_library_scores_vs_counted_rate.tsv | Counted division rate, 20-marker proliferation score and the 177- and 96-gene splicing scores for the 328 counted libraries (Figs 2, 6a and 8)                        |
| Supplementary Data 8 | SupplementaryData8_intervention_prediction.tsv        | Out-of-sample prediction of the intervention classes: contrasts, series, median change and                                                                            |

| File                  | Deposited as                                          | Contents                                                                                                                                                                                                                                                                              |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                                       | residual under each prediction scheme (Supplementary Fig. S4, Supplementary Results 5)                                                                                                                                                                                                |
| Supplementary Data 9  | SupplementaryData9_senescence_panels.tsv              | The seven senescence panels: gene counts, cell-cycle content, correlation with the counted division rate with and without cell-cycle genes, and the age effect before and after adjustment (Supplementary Fig. S6, Supplementary Table 7)                                             |
| Supplementary Data 10 | SupplementaryData10_methylation_tests.tsv             | The four preregistered methylation clock tests with their correlations, <i>P</i> values, BH-FDR and verdicts (Supplementary Table 8)                                                                                                                                                  |
| Supplementary Data 11 | SupplementaryData11_gene_vs_counted_rate.tsv          | Every expressed gene of the counted resource with its correlation to the counted division rate, its mean expression, and whether it belongs to the 177-gene set or to the twenty named factors; for the 177, the fold change per division per day with its <i>P</i> and FDR (Fig. 2c) |
| Supplementary Data 12 | SupplementaryData12_junction_psi_vs_proliferation.tsv | Every testable splicing event of the donor cohort with its inclusion level and its association with the proliferation score and                                                                                                                                                       |

| File                  | Deposited as                             | Contents                                                                                                                                                                                                                                                         |
|-----------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supplementary Data 13 | SupplementaryData13_gene_age_effects.tsv | <p>with donor age, before and after adjustment (Fig. 3)</p> <p>Each of the 177 genes in the donor cohort: the age effect in <math>\log_2</math> counts per million per decade with its <math>P</math> and FDR, before and after adjustment for proliferation</p> |